Abstract

A fixed array of physical qubits can be spent on unprotected circuit replicas with error mitigation, on surface-code logical qubits with analog rotation injection (STAR), or on fully fault-tolerant logical qubits fed by magic-state factories, and which of these buys the most useful computation depends on how each converts area and time into accuracy. In this work, we compare these architectures, together with a hybrid that injects only variable rotations, on one nearest-neighbor array with one week of execution, for two workloads: the random circuits of the QUOPS benchmark, whose rotation angles are of order one, and the central doublon density of an $L \times L$ Fermi–Hubbard lattice evolved for a time $\tau = L/4$, whose Trotter angles are small. A common accounting model optimizes each architecture's error allocation, code distance, and non-Clifford delivery, using measured routing, a validated polarization model, exact finite-lattice Trotter errors, and a circuit-level calibration of the surface-code law under uniform circuit noise ($\Lambda = 6.3$ at $p = 10^{-3}$) that also prices distillation factories. At a physical error of 10^{-3} on every operation, physical qubits score about 240 QUOPS and STAR saturates near 6×10^3, while full fault tolerance leads from 2.5×10^4 qubits and scores 4.8×10^6 with 10^6. On the Hubbard task, whose small angles favor analog injection, physical-qubit computation saturates at $\tau \approx 1.5$ and STAR and the hybrid at $\tau \approx 2.5$–3, while full fault tolerance is never behind from about 3.5×10^5 qubits and reaches $\tau = 8$ with 10^7. The T source is the largest delivery lever: at strict precision, distillation factories move the crossovers by more than an order of magnitude, whereas lifting STAR's delivery limits moves none, because STAR is limited by exposure. A controlled sweep of the injected angles gives STAR a saturation size $s \approx 0.41/(p\langle|\theta|\rangle)$. These results are conditional on declared delivery models, Trotter coefficients converged only near $\tau = 1$, and an assumed mitigation overhead for physical qubits. They are consistent with a simple exposure picture: physical qubits pay per gate, STAR per radian, and full fault tolerance per synthesized rotation with only a logarithmic precision overhead.

1 Introduction

A fixed array of physical qubits can be spent on many unprotected circuit replicas, on logical qubits whose non-Clifford rotations are injected in analog, or on fully fault-tolerant logical qubits fed by magic-state factories. Which of these buys the most useful quantum dynamics is not settled by any of the three in isolation; it depends on how each converts physical area and one week of wall-clock time into accuracy. Physical-qubit computation with error mitigation pays an exponential sampling cost in its accumulated physical error [1]. The space–time efficient analog rotation (STAR) architecture protects Clifford operations with the surface code but injects arbitrary rotations with an error proportional to their angle [2, 3]. Full surface-code fault tolerance synthesizes every rotation from high-fidelity T states [4, 5, 6], at a large fixed cost in area.

In this work, we compare these uses of the same nearest-neighbor array on two workloads chosen to separate the controlling mechanisms. The first is the QUOPS benchmark of Proctor *et al.* [7]: random circuits whose score is the size of the largest circuit a machine runs faithfully. Its rotation angles are of order one, and its size axis is nearly continuous. The second is an application with the opposite angle structure: the doublon density at the center of an $L \times L$ Fermi–Hubbard lattice after a quench from a Néel state, with evolution time $\tau = L/4$ growing with the lattice, whose product-formula rotations are small. For each workload we ask how large a computation each architecture completes given Q physical qubits, a physical error rate p, and one week, and which modeling choices decide the answer. We consider four arms: physical qubits with error mitigation (the physical arm); STAR; full fault tolerance (FT); and a hybrid, which injects variable rotations in analog but draws fixed-angle rotations from the same T-state supply as FT. Figure 3 shows the resource roles.

Figure 1 summarizes the first answer. Physical qubits score about 240 QUOPS at $p = 10^{-3}$ however large the device, since replicas cannot raise a polarization. STAR overtakes them with about 10^4 qubits, briefly leads, and then saturates near 6×10^3. Full fault tolerance begins last, leads from 2.5×10^4 qubits, and scores 4.8×10^6 with 10^6 qubits, about 800 times STAR.

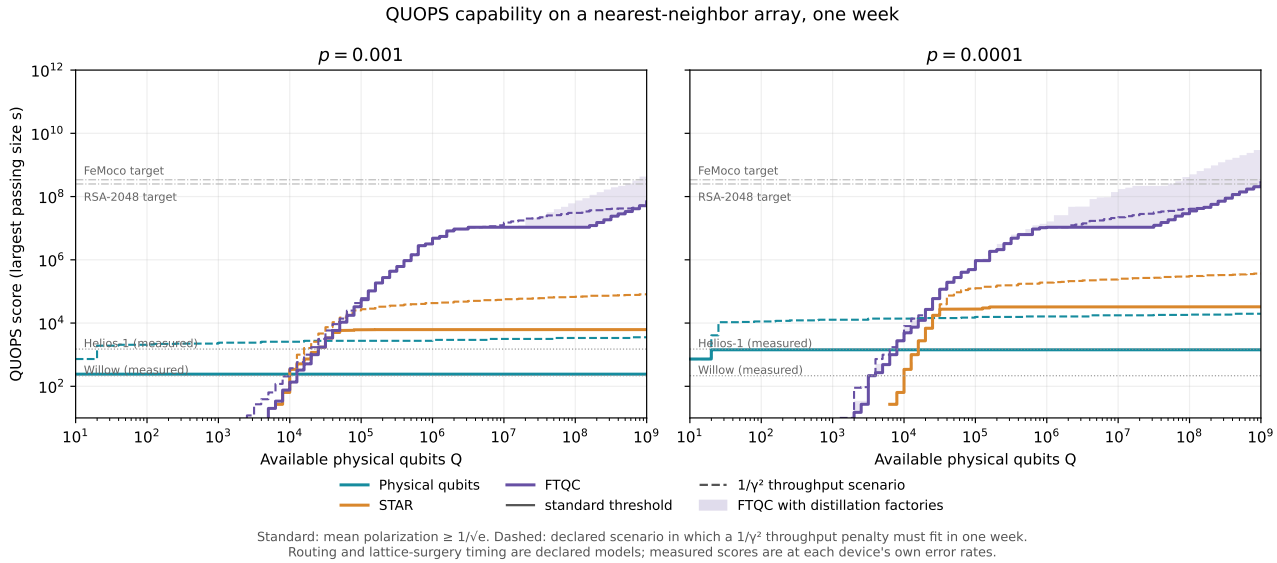


Figure 1: QUOPS score reachable within one week versus available physical qubits. Solid lines use the standard threshold $\bar{\gamma} \geq e^{-1/2}$; dashed lines show a declared scenario in which a $1/\bar{\gamma}^2$ throughput penalty must fit in the week. The shaded band adds the distillation factories of Ref. [8], re-optimized under the calibrated surface-code law, to full fault tolerance's source menu. Horizontal lines give the measured Willow and Helios-1 scores, each at its own error rates and connectivity, and the paper's RSA-2048 and FeMoco targets. Routing is measured; lattice-surgery timing is calibrated for isolated gadgets and declared for congestion.

Figure 2 gives the second. The Hubbard task is the one on which STAR should do best, and it does: the analog arms stay within a few lattice sites of full fault tolerance up to a few times 10^5 qubits and saturate at $\tau \approx 2.5$–3.5, while physical-qubit computation stops near $\tau = 1.5$. Full fault tolerance is never behind them from 3×10^5 qubits and keeps scaling to the sampled ceiling. Both orderings are consistent with one law of the exposure variables introduced below. Physical qubits pay per gate, STAR per radian, and full fault tolerance per synthesized rotation with only a logarithmic precision overhead. The two workloads differ most in how fast the radians grow, although they also differ in success criterion, connectivity demand, and the repeated angles that full fault tolerance can batch. A controlled sweep of the angle distribution under one success criterion (Sec. 5.2) isolates the angle and confirms the per-radian law.

The remainder of this paper builds these figures from their parts. Section 2 defines both tasks and the accounting shared by all arms. Section 3 describes the Trotter algorithms and the measured step-count law for the Hubbard task. Section 4 gives each architecture's cost model, with calibrations collected in the appendices. Section 5 reports capabilities and crossovers for both workloads and identifies which modeling choices move the leader. Section 6 states what controls the scaling, what remains conditional, and which calculations could change the conclusions.

2 Tasks and accounting

2.1 Random circuits

A QUOPS circuit of width w alternates a CNOT layer on $\lfloor w/2 \rfloor$ uniformly random pairs with a layer of rotations $e^{-i\theta P/2}$, where $P \in \{X, Y, Z\}$ and θ is uniform on $[0, 2\pi)$ [7]. Its size s counts one per

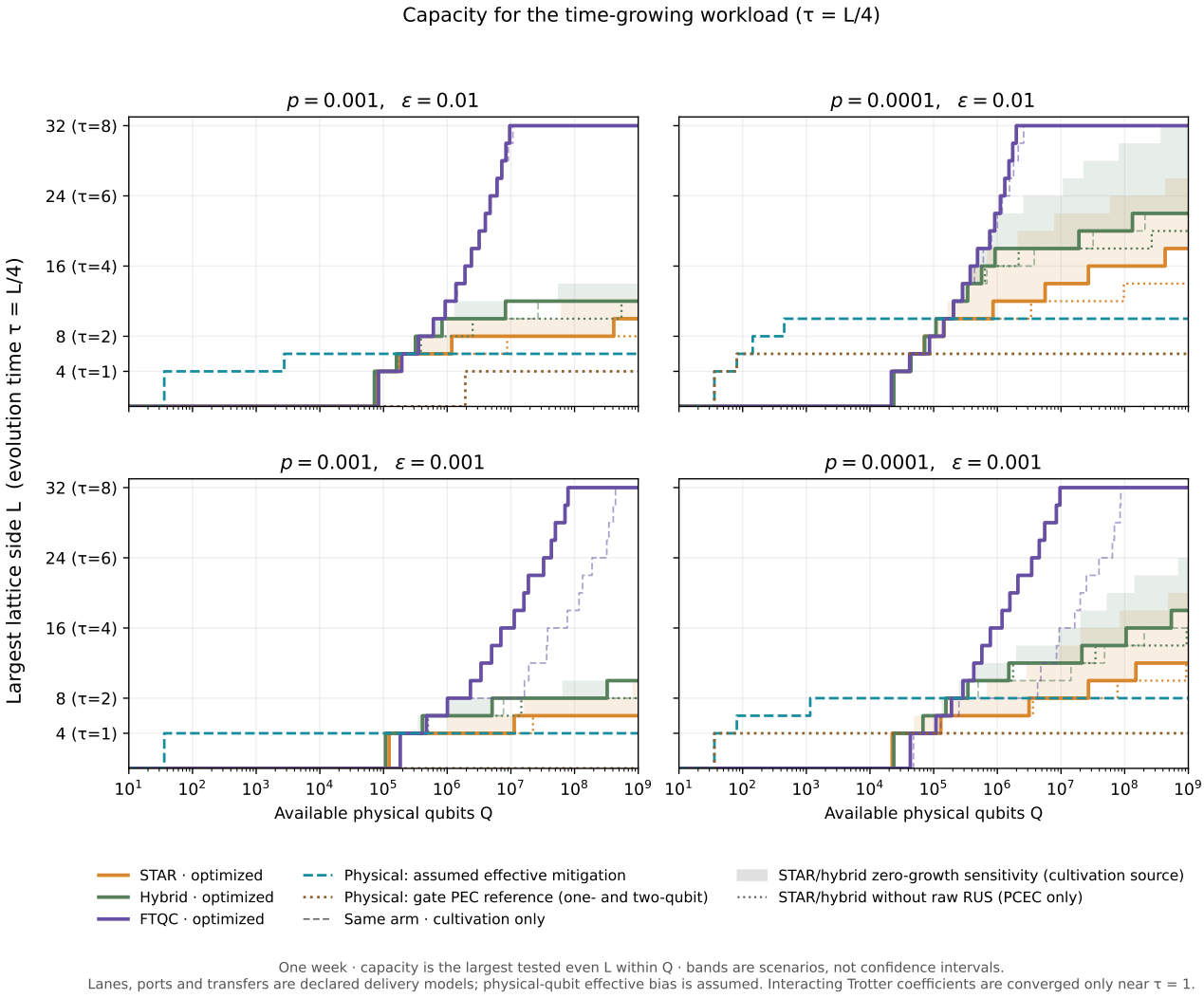


Figure 2: Largest lattice side L, and evolution time $\tau = L/4$, reachable within one week as a function of available physical qubits. Solid curves use the fully optimized model of Secs. 2–4 with the circuit-level surface-code law of Sec. 4.2, with T states from cultivation or distillation factories; thin dashed curves of the same color restrict the T source to cultivation, and dotted STAR and hybrid curves remove raw repeat-until-success, bounding the case in which its coherent cancellation fails. Bands give the STAR and hybrid zero-growth sensitivity, computed with cultivation. Physical-qubit curves assume an effective mitigation overhead or apply probabilistic error cancellation of every one- and two-qubit gate. $L = 32$ is the sampled ceiling. Delivery models are declared, not compiled, and interacting Trotter coefficients are converged only near $\tau = 1$.

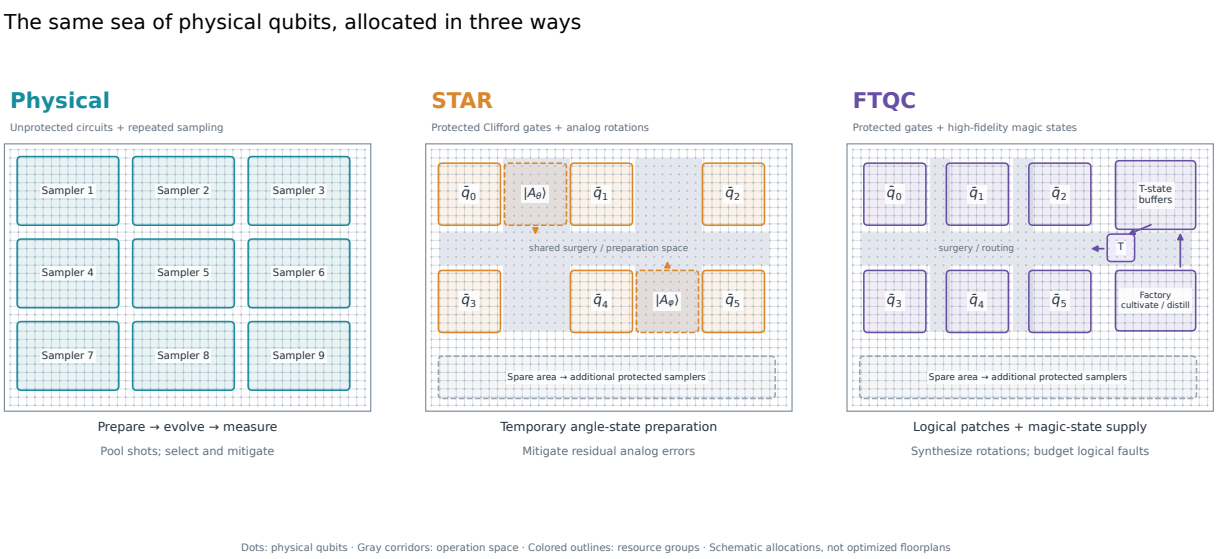


Figure 3: Three allocations of a common array of physical qubits. Physical-qubit computation uses many unprotected circuit replicas. STAR and FT share protected data patches and lattice surgery. Dashed orange patches show temporary angle-state preparation in STAR surgery ancillas [3]; FT supplies high-fidelity T states to a synthesized sequence of logical operations. The drawings show resource roles, not numerical floorplans.

rotation and two per CNOT, so a size-s circuit contains about $s/2$ rotations. A shape passes when the mean process polarization satisfies $\bar{\gamma} \geq e^{-1/2}$, and the score is the largest passing s with $w^2 \leq s \leq w^3$. The polarization is the product of each operation's process infidelity ϵ_i embedded in the full register,

$$\bar{\gamma} \simeq \prod_i \left(1 - \frac{4^w}{4^w - 1}\,\epsilon_i\right), \tag{1}$$

which reproduces exact noisy simulations of $w = 2$–8 random circuits to 0.2%, whereas the product of local gate polarizations is up to 24% pessimistic (Appendix F). For every shape and arm we compute the fewest physical qubits that pass within one week, including a mirror-circuit test of 10^4 executions at twice the forward size; the paper prescribes no shot count, and its experiments used 2×10^3–2×10^5. We also show a declared scenario in which the paper's approximate error-mitigation throughput penalty, $1/\bar{\gamma}^{\,2}$, must fit in the week. It is not the sample complexity of certifying a polarization, which would need an estimator and a confidence calculation.

2.2 Hubbard dynamics

The problem has $N = L^2$ sites, $2N$ spin modes, and open boundaries. In units of the hopping J,

$$H/J = -\sum_{\langle ij\rangle,\sigma} (c^\dagger_{i\sigma} c_{j\sigma} + \text{h.c.}) + u \sum_i (n_{i\uparrow} - \tfrac{1}{2})(n_{i\downarrow} - \tfrac{1}{2}), \qquad u = U/J = 4, \tag{2}$$

where the symmetric onsite form differs from $U n_\uparrow n_\downarrow$ only by a known phase at fixed particle number. We prepare the half-filled Néel product state and estimate the central doublon density $D = \langle n_{i\uparrow} n_{i\downarrow}\rangle$ at time $\tau = Jt/\hbar$.

We tie the evolution time to the lattice, $\tau = L/4$. At fixed τ, a local observable stops depending on L once the lattice contains its light cone; for free fermions ($u = 0$) the central doublon at $\tau = 1$

differs from its $L = 60$ value by 2.4×10^{-3} at $L = 6$ and by 2.4×10^{-4} at $L = 8$. A capacity measured in L at fixed time therefore stops measuring physics beyond a modest size, whereas the time-growing workload keeps the relevant region and the lattice growing together. The fixed-time comparison is retained as an implementation benchmark in Appendix G. The free-fermion numbers motivate this choice but do not bound the interacting dynamics.

Quantity	Value
Connectivity	Square array, nearest neighbors
Gate / measurement / reset	10 / 100 / 100 ns
Physical error	$p = 10^{-3}$ on every gate, idle, reset, and measurement; $p = 10^{-4}$ as a sensitivity
Syndrome cycle / feedforward	240 / 200 ns
Runtime cutoff	one week, $B = 604800$ s
Accuracy	absolute $\epsilon = 0.01$ and $\epsilon = 0.001$ on D, 95% confidence
Rotation convention	$U_P(\theta) = e^{-i\theta P}$

Table 1: Fixed hardware and task parameters.

2.3 Accuracy contract for the Hubbard task

The estimate must satisfy $|\widehat{D} - D| \le \epsilon$, where the error is split among independent contributions,

$$\epsilon = \epsilon_{\text{stat}} + \epsilon_{\text{Trot}} + \epsilon_{\text{bias}} + \epsilon_{\text{inst}} + \epsilon_{\text{logical}} + \epsilon_{\text{synth}} + \epsilon_{\text{magic}}, \tag{3}$$

where ϵ_{stat} is the statistical half-width, ϵ_{Trot} the product-formula error, ϵ_{bias} a reserve for residual mitigation bias (0.1ϵ for the physical, STAR, and hybrid arms), ϵ_{inst} the coherent residue and retry tail of analog injection, $\epsilon_{\text{logical}}$ the logical-fault bias, and ϵ_{synth} and ϵ_{magic} the rotation synthesis and magic-state errors. Biases are added linearly, which is conservative. The optimizer chooses the split, together with the step count, code distance, and delivery, separately for every arm and lattice.

2.4 Sampling, replicas, and capacity

Let a mitigated estimator have per-attempt variance multiplier e^{χ}, where χ is its exposure. With the doublon variance planned at $D_*(1 - D_*)$ for $D_* = 0.1$, the ideal shot count and the qubits needed to finish within the cutoff are

$$N_0 = \left\lceil \frac{z^2 D_*(1 - D_*)}{\epsilon_{\text{stat}}^2} \right\rceil, \qquad Q_{\min} = q_a \left\lceil \frac{N_0\, e^{\chi}\, t_a}{B} \right\rceil, \tag{4}$$

where $z = 1.96$, q_a is the area of one complete sampler, and t_a its attempt time; replicas run in parallel, and a sampler whose single attempt exceeds B is excluded. At $\epsilon_{\text{stat}} = 0.004$ this gives $N_0 \approx 2.2 \times 10^4$. Every arm pays Eq. (4); FT has $\chi = 0$ but a large q_a. Inverting it gives the exposure an architecture can afford,

$$\chi \lesssim \log \frac{QB}{q_a N_0 t_a}, \tag{5}$$

so additional hardware or runtime buys exposure only logarithmically. We report the capacity $L_{\max}(Q)$, the largest tested lattice with $Q_{\min} \le Q$; it is a discrete upper envelope over every even L from 4 to 32 and is not interpolated.

3 Algorithm

3.1 Two second-order product formulas

Both formulas use the Jordan–Wigner encoding on a snake and drop the single-Z onsite terms, which contribute a known phase at fixed particle number. With $E = 2L(L-1)$ bonds, the remaining Hamiltonian has

$$K = 4E + N = 9N - 8L, \qquad \lambda = \sum_{\ell} |h_\ell| = 2E + \frac{|u|N}{4} = 5N - 4L, \tag{6}$$

terms and one-norm, respectively. The termwise Strang formula applies every term as a Pauli rotation. Merging adjacent identical rotations gives

$$R_{\text{term}} = r(2K-2) + 1, \qquad \Theta_{\text{term}} = \tau\lambda, \tag{7}$$

where r is the step count, R the number of rotations, and Θ their total angle. Notice that Θ does not depend on r: refining the step multiplies the number of rotations but not their total angle. This identity is what makes small-angle injection attractive.

Campbell's plaquette formula [9] partitions hopping into disjoint four-site tiles, whose evolution needs two variable rotations and eight fixed T gates per spin. Adapted to open boundaries with $P = L^2/4$ pink plaquettes, $G = (L/2-1)^2$ gold plaquettes, and $B = 2(L-2)$ isolated boundary bonds,

$$R_{\text{plaq}} = N(r+1) + 4r(2P + G + B), \qquad T_{\text{fixed}} = 16r(2P+G), \tag{8}$$

$$\Theta_{\text{dyn}} = \tau\left[\frac{|u|N}{4} + 4(P+G) + 2B\right]. \tag{9}$$

At $L = 4$, $\tau = 1$, and $r = 4$ the plaquette circuit has 288 variable rotations of total angle 44 and 576 fixed T gates, against 889 rotations of total angle 64 for termwise Strang. STAR uses the termwise formula, since fixed $\pi/8$ rotations are expensive to inject in analog; FT and the hybrid use plaquettes.

3.2 The step-count law

For a local observable at fixed τ, the Trotter error does not grow with the lattice once the lattice contains the observable's causal region [10]. We therefore use a law without explicit size dependence,

$$\epsilon_{\text{Trot}} = \frac{C(\tau)}{r^2}, \qquad r \geq 8\tau, \tag{10}$$

where $C(\tau)$ is 1.1 times the running maximum through τ of the measured $|D_r - D|r^2$ over exact 2×2, 3×3, 4×3, and 4×4 dynamics. At $\tau = 1$, $C = 0.29$ (plaquette) and 0.21 (termwise); by $\tau = 4$ they grow to 5.8 and 3.4. The measured errors are cleanly quadratic: on the 4×4 plaquette patch at $\tau = 1$ they are 4.14×10^{-3}, 1.03×10^{-3}, and 2.56×10^{-4} at $r = 8$, 16, and 32. The plaquette coefficient is nearly converged across patches, 0.22–0.26 at $\tau = 1$ from 2×2 to 4×4. The termwise coefficient grows from 0.077 on 3×3 to 0.19 on 4×4, but exact dynamics on strips up to 8×2 and 6×3 show that this growth stops: at $\tau = 1$ it reaches 0.18–0.19 by length 5 and stays there, and widths 2, 3, and 4 agree within 5% (Appendix A.3). At $\tau = 2$ it converges in length but still rises slowly with width, from 0.68 to 0.78, below the planning value of 0.86. At $U = 0$, where both circuits are Gaussian and can be simulated exactly up to 24×24, the coefficient likewise stops changing once $L \gtrsim 4\tau + 2$ (Appendix A.4). Beyond $\tau \approx 1$ the two-dimensional value therefore remains a transferred assumption, now bounded by every measured patch. Appendix A describes the matrix-free calculations, the nested-patch checks, and the transfer to larger lattices, which remains an assumption.

3.3 Richardson extrapolation as an option

Because the error is quadratic, two branches with step counts r and $2r$ and coefficients $c = (-\frac{1}{3}, \frac{4}{3})$ cancel its leading term [11, 12]. The residual is measured as K_4/r^4; on the 4×4 plaquette patch $|4E(2r) - E(r)|r^4/3 = 0.0514$ at both $r = 8$ and 16. The extrapolation is not free. Optimal allocation of shots between branches with variance multipliers F_i and attempt times t_i costs

$$T_{\text{sample}} \propto \Big(\sum_i |c_i| \sqrt{F_i t_i} \Big)^2, \tag{11}$$

which with equal F and $t_2 = 2t_1$ is $(\frac{1}{3} + \frac{4}{3}\sqrt{2})^2 = 4.92$ times the base branch. Branch biases add with weights $|c_i|$, so the other allowances are divided by $\|c\|_1 = 5/3$. The optimizer uses it only where the freed Trotter allowance buys more than this cost, which happens at $\epsilon = 0.001$.

4 Architectures

Each architecture below is costed by the same design search, which chooses its step count, error allocation, code distance, and delivery; Appendix H lists the full menu, the declared inputs, and the calibrated constants.

4.1 Physical qubits with error mitigation

The physical circuit is a fermionic-swap network on the nearest-neighbor array [13]: the Jordan–Wigner encoding, with fermionic swaps that bring each hopping pair together so that every gate acts on neighboring qubits (Appendix E.1). We anchor its size to the published Willow compilation of the same model at 6×6 and three steps [14],

$$G = 4372 \, \frac{r}{3} \left(\frac{L}{6} \right)^3, \qquad t_a = t_{\text{reset}} + 294 \, \frac{r}{3} \, \frac{L}{6} \, t_g + t_{\text{meas}}, \tag{12}$$

where G is the two-qubit gate count and $t_g = 10$ ns; a replica occupies the smallest square holding $2N+2$ qubits. Two estimators bracket the arm. The reference is probabilistic error cancellation (PEC) of every gate. The depolarizing inverse of a two-qubit gate has $\gamma_2 = 1 + 2p/s_2$ with $s_2 = 1 - 16p/15$, and the compiled network has 0.8 one-qubit gates per two-qubit gate, each with the same error p and the same inverse cost $\simeq 4p$, so that $\chi_{\text{PEC}} \simeq (4 + 0.8 \cdot 4) pG \approx 7pG$; idle errors are not charged. The second line assumes an effective mitigation overhead $\chi_{\text{eff}} = \kappa pG$ with $\kappa = 1.05$, chosen so that the Willow circuit costs a factor 100 in variance; it is not fitted. Measured attenuation in the Phasecraft and Willow data points to $\kappa \approx 0.2$–0.4, but neither experiment establishes our absolute accuracy with controlled bias. For this reason we also solve the inverse problem: the variance and remaining offset that any physical estimator would need to match the encoded arms (Appendix E).

The physical arm meets its wall abruptly. Holding the Trotter error at a fixed absolute accuracy needs about eight steps per unit time, so $r \propto \tau \propto L$ and $G \propto rL^3$ give an exposure $\chi \propto L^4$ for the time-growing workload, while Eq. (5) lets the affordable exposure grow only as $\log Q$. The capacity therefore scales as $L_{\max} \propto (\log Q)^{1/4}$. Under the assumed effective overhead at $p = 10^{-3}$ and $\epsilon = 0.01$, $L = 6$ ($\tau = 1.5$, twelve steps, 1.7×10^4 gates) needs about 3×10^3 qubits, while $L = 8$ ($\tau = 2$) needs 10^{21}; six decades of hardware buy about 20% in lattice side. The published Willow experiment reached 6×6 with one to three Trotter steps and at most 4372 two-qubit gates [14]. At 4×4 and three steps its processed local doublon matches continuous interacting dynamics to 0.0103 RMS, at our accuracy target (Sec. 5.4); a 6×6 lattice at $\tau = 1.5$ and $\epsilon = 0.01$ needs four times their largest circuit.

For random circuits the physical arm must also route arbitrary pairs on the nearest-neighbor array. We use laws measured on 900 routed QUOPS circuits of widths 4–144 (Appendix F): about $0.35w^{1.59}$

fused two-qubit gates per layer with Cirq's router, or depth growing only as $w^{0.62}$ with a constructive sorting network, whichever loses less polarization at each width. The noise is the uniform model of the surface-code calibration: depolarizing error p per two-qubit gate, per one-qubit gate, and per idle 10 ns slot, each entering Eq. (1) with process infidelity equal to its total Pauli error probability. Without mitigation, replicas only shorten the mirror test.

4.2 Encoded protection

STAR, FT, and the hybrid store logical qubits in surface-code patches of $2d^2$ physical qubits and perform Clifford operations by lattice surgery. Each patch fails per syndrome round with probability

$$p_L(d) = A\,\Lambda^{-(d+1)/2}, \qquad \Lambda = p_{\mathrm{th}}/p. \tag{13}$$

A circuit-level memory calibration under standard uniform noise (depolarizing error p after every gate and on every idle qubit in every layer, and flips of probability p on resets and measurements), using Stim and PyMatching [15, 16] with 9.6×10^7 shots at $d = 3$–9, gives $\Lambda = 6.30$ (95% interval 6.18–6.43) and $A = 0.103$ at $p = 10^{-3}$, and $\Lambda = 44$ at $p = 10^{-4}$ (Appendix B). This is the noise model of the cultivation and distillation estimates we use. The widely used reference $A = 0.1$, $\Lambda = 10$ is optimistic by 6–8 in code distance at the operating points of this study. The same law serves every encoded arm, so it favors none of them against the others.

For V patch-rounds per attempt, the logical-fault bias is at most $V p_L(d)$. Under PEC of the analog rotations, a fault's bias can be amplified by up to $e^{\chi/2}$, so the smallest admissible distance satisfies

$$V\,p_L(d) \le \epsilon_{\mathrm{logical}}\,e^{-\chi/2}. \tag{14}$$

Because V itself depends on d through the attempt time, the distance is found self-consistently. Faults are budgeted per attempt; the campaign is not required to contain zero logical faults. On random circuits, each layer of long-range CNOTs is assumed to take two code clocks of lattice surgery in the fast data block without congestion, a declared model. An isolated circuit-level calibration under the same noise model supports its two ingredients for adjacent patches (Appendix B): a lattice-surgery CNOT with d merge rounds fails at 0.73–1.15 times the memory law evaluated on its space-time volume, and it takes $2d$ clocks plus 300 ns.

4.3 STAR

STAR teleports each rotation from an angle-specific resource state. A failed teleportation applies the rotation with the opposite sign and is corrected by injecting twice the angle, repeating until success (RUS) [2, 3]; Figure 4 shows the protocol. We evaluate the transversal multi-rotation preparation of Ref. [3] exactly at every workload angle: a $k = (d_i+1)/2$ factor preparation at injection distance d_i has undetected generator faults with probability $p/15$ per pair factor and $p/30$ for the final factor, which fixes both the preparation acceptance and the accepted channel. The resulting rotation multiplies off-diagonal elements by c_j, and PEC of its dephasing costs

$$\chi_{\mathrm{STAR}} = -2\sum_j \log(\mathrm{Re}\,c_j) \simeq 4\alpha p\,\Theta_{\mathrm{term}}, \tag{15}$$

where the sum runs over rotations with multiplicity and α, of order 1–3, is an effective prefactor that depends on angle and injection distance. The coherent residue $|\,\mathrm{Im}\,c_j|$ and the finite RUS tail are charged to ϵ_{inst}. Injection seeds $d_i = 5$, 9, and 25 grow to receivers of distance 13 or 25. Their growth error is the upper limit from 2.4×10^7 decoded growth shots, composed as an independent channel; most campaigns observed no failures, so this is a bound rather than a measured rate (Appendix C).

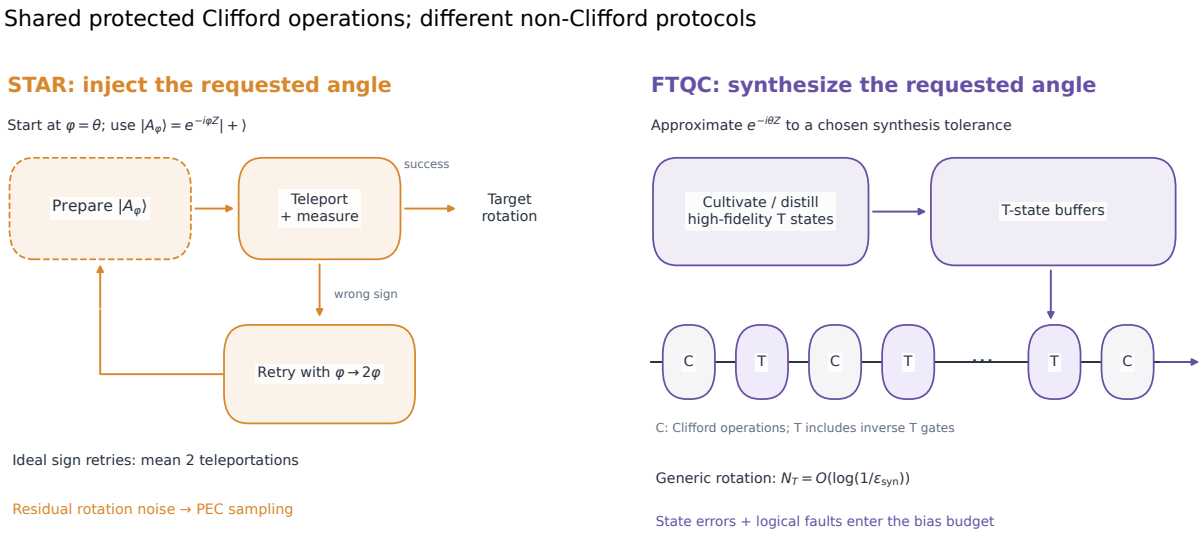


Figure 4: Two implementations of the same target rotation, in the convention $e^{-i\theta Z}$. STAR prepares an angle-specific resource state and uses teleportation with angle-doubling corrections [2, 3]. Ideal sign outcomes require two teleportations on average; this does not include rejected preparation attempts or their cost. FTQC consumes high-fidelity T states in a Clifford+T approximation, whose generic T count grows logarithmically with inverse synthesis tolerance [4]. The hybrid chooses between these paths by operation class. These are protocol flows, not schedules with equal durations or calibrated factory footprints.

Two refinements of the protocol are options of the optimizer. Probabilistic coherent error cancellation (PCEC) removes the coherent part of each trial and roughly doubles its dephasing [3]. Without it, averaging over RUS branches already cancels the coherent term at first order in our instrument, halving the exposure. Toshio *et al.* report the opposite ordering for the branch-averaged channel; within their own model the difference is the sign of a failed trial's coherent error, and the physical sign gives our result (Appendix D). Rotations can also be injected in parallel lanes. Each lane occupies six tiles; disjoint Jordan–Wigner supports limit concurrent termwise rotations to at most L; and more than four lanes use the fast data block. The attempt time is the analog service time divided by the number of lanes, plus surgery and handoff. For random circuits the injected angle is $\theta/2$ reduced modulo $\pi/4$ by Clifford corrections, which is uniform on $[-\pi/8, \pi/8]$; we average the instrument over that distribution. Its mean magnitude, $\pi/16$, is roughly seven times the Trotter angles of the Hubbard task.

4.4 Full fault tolerance

FT synthesizes each variable rotation to operator error ζ with

$$n_T = \lceil 3.02 \log_2(1/\zeta) + 0.77 \rceil , \qquad \zeta = \frac{\epsilon_{\text{synth}}}{2R}, \tag{16}$$

T gates [4]. Because the plaquette angles are shared by every tile of a stage, Hamming-weight phasing can replace a batch of b equal rotations by $\lfloor \log_2 b \rfloor + 1$ rotations and $b - \text{popcount}(b)$ Toffolis, with that workspace charged [9]. Magic states come from cultivation [6]: 1024 qubits and about 100 cycles per state at error 2×10^{-9}, the published estimate at $p = 10^{-3}$, with an optional 15-to-1 cleanup stage of twelve tiles and output error $35q^3$. Each factory costs $1024 + 2d^2$ qubits, or $1024 + 13 \cdot 2d^2$ with cleanup. An exact calculation of the same circuit gives 3.3×10^{-9} before its final growth stage [17], so the published value is optimistic; Sec. 5.7 repeats the optimization at 4×10^{-9}. The optimizer may instead use the distillation factories of Ref. [8]: one- and two-level 15-to-1 and 20-to-4 protocols whose distances we re-optimized with the paper's own factory model after replacing its logical-error law, $0.1(100p)^{(d+1)/2}$, by our calibrated one. At the published output errors these need about 50% more qubits at $p = 10^{-3}$ than the published two-level designs, whose output errors under our law are 30 to 4×10^4 times larger. Each factory adds one delivery tile, and the menu keeps 24 designs at $p = 10^{-3}$ and 32 at 10^{-4} (Appendix B).

Supply and consumption run concurrently. A synthesized rotation is a serial T sequence, whereas the rotations of one plaquette stage are independent, so with P consumption ports the serial T depth is

$$N_T^{\text{serial}} = n_T \sum_s m_s \left\lceil \frac{R_s}{P} \right\rceil + \frac{N_T - n_T R}{P}, \tag{17}$$

where stage s is repeated m_s times with R_s synthesized rotations and N_T is the total T count. With F factories of service time s_F, the attempt time is

$$t_a = \Big[2d + \max\Big(d\big(c_T N_T^{\text{serial}} + W/P\big),\ \lceil N_T/F \rceil s_F \Big) \Big] t_c + N_T^{\text{serial}} t_{\text{ff}} + t_{\text{reset}} + t_{\text{meas}}, \tag{18}$$

where c_T is 9 clocks per T for the compact block with one port and one clock for the fast block, W counts two Clifford clocks per rotation and per Toffoli, $t_c = 240$ ns, and t_{ff} is the feedforward latency. The optimizer chooses block, ports, factories, batch size, cleanup, and distance together; each extra port costs two tiles. The cultivated state is transferred from its distance-15 patch into the data distance at a cost of $2\max(d, 15)$ cycles, so the data distance may be smaller than 15. For random circuits we follow the paper and charge each synthesized rotation its squared diamond-distance error, ζ^2, in Eq. (1); random angles admit no Hamming-weight batching.

4.5 The hybrid

The hybrid runs the plaquette circuit, injects its variable rotations as in STAR, and draws its fixed $\pi/8$ rotations from the FT magic-state supply. It therefore pays STAR's exposure only on Θ_{dyn} of Eq. (9), without synthesizing any rotation, at the cost of factory area and a magic-state error amplified by $e^{\chi/2}$. The distilled $\pi/8$ rotations are the source of its advantage: injecting them through the STAR instrument instead makes the plaquette circuit worse than STAR's termwise circuit beyond $L = 8$, because they carry about eleven times the circuit's dynamic angle (Sec. 5.7). A proposed extra protection step pays at fixed shot budget when

$$\Delta \log t_a + \Delta\chi - \Delta \log m < 0, \tag{19}$$

where m is the number of replicas; the hybrid is the simplest instance of this operation-dependent protection.

5 Results

5.1 Random circuits

Figure 1 shows the capability of each arm on random circuits. At $p = 10^{-3}$ and under the standard threshold, physical-qubit computation scores about 240 whatever the device size, since extra replicas cannot raise a polarization. A reimplementation by another workshop group obtains about 150 at the same p, mostly because it charges readout error, which the mirror test excludes from the polarization. Willow measured 216 at its own error rates. STAR overtakes the physical arm at 10^4 qubits and leads alone between 10^4 and 2×10^4, where full fault tolerance is still limited by its fixed area, but saturates at 6.2×10^3 (3.6×10^3 without raw repeat-until-success), because each random rotation injects an angle of average magnitude $\pi/16$, roughly seven times the Trotter angles of the Hubbard task. Full fault tolerance leads for good from 2.5×10^4 qubits, reaches 10^4 QUOPS with 5×10^4 qubits, and scores 4.8×10^6 at 10^6 qubits, about 800 times STAR; the exact cultivation error of 4×10^{-9} changes no capability. On the Hubbard task, by contrast, the analog arms stayed within a few lattice sites of full fault tolerance up to a few times 10^5 qubits. The difference is the one the exposure picture predicts: STAR pays per radian, and here the radians are large.

Full fault tolerance then plateaus near 1.1×10^7 from about 3×10^6 to 10^8 qubits. Beyond that size a circuit needs more than about 2×10^8 T states, which exceeds what cultivated states of error 2×10^{-9} can supply within the polarization budget, and every factory must add a 15-to-1 cleanup stage that is roughly seven times larger and fifteen times slower. At 10^9 qubits the score reaches 6.8×10^7, short of the paper's RSA-2048 and FeMoco targets of 2.5–3.4 $\times 10^8$; at $p = 10^{-4}$ it reaches them with about 10^9 qubits. The plateau is a feature of source quality, not of full fault tolerance. When the distillation factories of Ref. [8], re-optimized under the calibrated surface-code law, join cultivation in the source menu, the plateau disappears: the score reaches the RSA-2048 target with 5×10^8 qubits at $p = 10^{-3}$, and at $p = 10^{-4}$ with 5×10^7 qubits instead of 10^9. The published factories, priced with the reference law, give 3.2×10^8 and 1.3×10^7. Under the $1/\bar{\gamma}^2$ throughput scenario the physical arm reaches about 3.6×10^3 and STAR about 8×10^4 at 10^9 qubits, without changing the ordering. The mirror test's shot count matters only where logical shots are slow: with Willow's 2×10^5 executions instead of 10^4, full fault tolerance beyond $s \approx 10^7$ needs roughly an order of magnitude more qubits.

For the shapes of the paper's planar projection, our full fault tolerance needs 3.8–5.0 times more physical qubits than reported there, for example 4.7×10^4 rather than 10^4 for $w = 21$ and $s = 9240$. Our calibrated $\Lambda = 6.3$ raises the code distance relative to the reference law, and the fast data block and 1024-qubit cultivation sources add area; we report the ratio rather than tune toward it.

5.2 A controlled angle sweep

The two workloads differ in more than their angles. To isolate the angle, we keep the random-circuit family, its success criterion, routing, and delivery fixed, and draw STAR's injected angles uniformly on $[-a, a]$ for a from 0.003 to $\pi/8$. The physical and fault-tolerant arms do not depend on the angles, so only STAR's exposure changes. Figure 5 shows the result. Below about 0.03 rad, STAR's saturation size follows

$$s_{\mathrm{sat}} \simeq \frac{0.41}{p\,\langle|\theta|\rangle}, \tag{20}$$

where $\langle|\theta|\rangle$ is the mean injected angle; the product $s_{\mathrm{sat}}\,p\,\langle|\theta|\rangle$ stays between 0.40 and 0.42 across two decades of angle at both $p = 10^{-3}$ and 10^{-4}. At larger angles the injected error grows more slowly than linearly, and at the QUOPS angles STAR does up to three times better than Eq. (20).

The angle sets the height of STAR's plateau, not the crossover. Shrinking the angles to the Hubbard range raises the plateau about forty times, yet full fault tolerance still overtakes STAR for good at 6×10^4 qubits instead of 2.5×10^4. Full fault tolerance wins by continuing past the plateau, and the crossover sits where its fixed area is paid. That is why the Hubbard task, with its small angles, keeps the analog arms competitive only up to a few times 10^5 qubits.

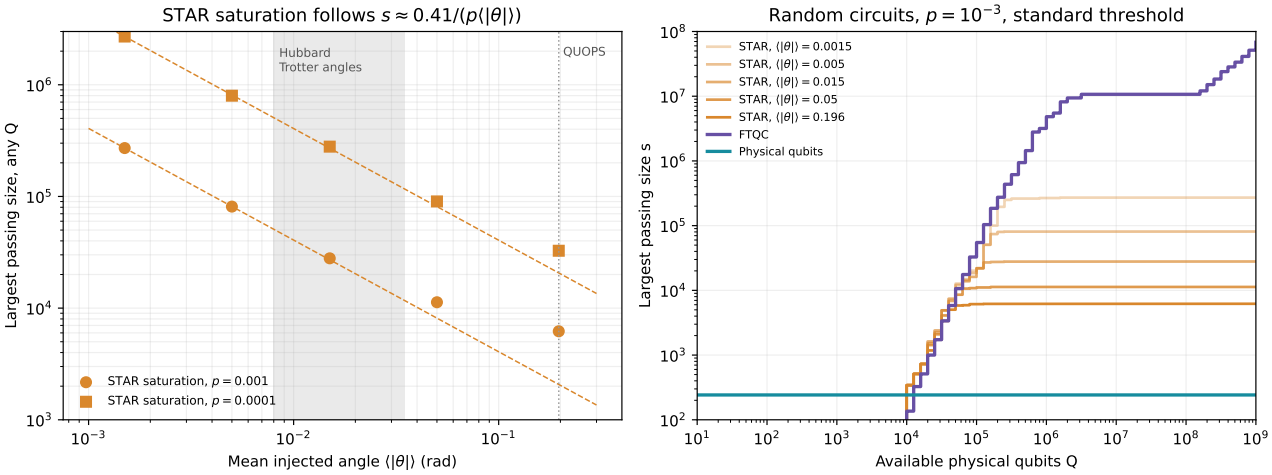


Figure 5: Angle sweep on random circuits under the standard threshold. Left: STAR's saturation size against the mean injected angle, with the law $s \approx 0.41/(p\langle|\theta|\rangle)$ (dashed); the shaded band marks the mean Trotter angles of the Hubbard task. Right: capability at $p = 10^{-3}$ for each angle scale, with full fault tolerance and physical qubits, which do not depend on the angles.

5.3 Hubbard capacity and crossovers

Figure 2 and Table 2 give the capacity of each arm for the time-growing Hubbard workload. At $p = 10^{-3}$ and $\epsilon = 0.01$, physical-qubit computation reaches $L = 6$ ($\tau = 1.5$) with about 3×10^3 qubits and stops there. Encoded computation overtakes it at about 3.2×10^5 physical qubits. Full fault tolerance is never behind the analog arms from 3.5×10^5 qubits, strictly ahead from 9.3×10^5, and reaches the $\tau = 8$ ceiling at 10^7, while STAR saturates at $L = 10$ ($\tau = 2.5$) and the hybrid at $L = 12$ ($\tau = 3$). At $\epsilon = 0.001$ full fault tolerance is never behind from 4.7×10^5 qubits, strictly ahead from 10^6, and reaches the ceiling with 8×10^7; with the exact cultivation error the first two points move to 7.7×10^5 and 1.4×10^6, the only entries of Table 2 that change appreciably. At $p = 10^{-4}$ every arm reaches further: the analog arms saturate at $\tau = 4.5$–5.5 for $\epsilon = 0.01$, and full fault tolerance reaches

the ceiling at 2×10^6 qubits. Mainly strict precision depends on the T source; with cultivation alone the $\tau = 8$ ceiling at $p = 10^{-3}$ and $\epsilon = 0.001$ costs 4.4×10^8 qubits (Sec. 5.7).

p	ϵ	Encoded > physical (PEC / eff.)	FT never behind	STAR / hybrid	FT at $\tau = 8$
10^{-3}	0.01	0.7×10^5 / 3.2×10^5	3.5×10^5	$\tau = 2.5$ / 3	9.6×10^6
10^{-3}	0.001	1.1×10^5 / 4.1×10^5	4.7×10^5	$\tau = 1.5$ / 2.5	7.9×10^7
10^{-4}	0.01	0.7×10^5 / 2.0×10^5	2.1×10^5	$\tau = 4.5$ / 5.5	2.0×10^6
10^{-4}	0.001	0.7×10^5 / 2.9×10^5	1.9×10^5	$\tau = 3$ / 4.5	9.6×10^6

Table 2: Physical qubits at which each transition occurs for the time-growing workload, with all optimizer options, uniform noise, the calibrated surface-code law, and T states from cultivation or distillation factories. "Encoded > physical" compares with PEC of every gate, which has a bound behind it, and with the assumed effective mitigation line. "FT never behind" is the point beyond which no analog arm exceeds FT at any larger budget; ties are allowed. "STAR / hybrid" gives the evolution time at which each saturates. Lattices are every even L from 4 to 32, and budgets are sampled on a logarithmic grid.

For the time-growing workload, full fault tolerance leads alone in every sampled cell with at least 10^7 qubits. Below that, the hybrid ties it at $p = 10^{-3}$, $\epsilon = 0.001$, and 10^6 qubits; at 10^5 qubits the analog arms tie it or, at $p = 10^{-4}$ and $\epsilon = 0.001$, the hybrid leads alone. STAR leads alone only in slivers of qubit budget, for example between 2.2 and 2.3×10^4 qubits at $p = 10^{-4}$ and $\epsilon = 0.001$. With cultivation alone the hybrid also ties full fault tolerance at $p = 10^{-3}$, $\epsilon = 0.001$, and 10^7 qubits and leads alone at $p = 10^{-4}$, $\epsilon = 0.001$, and 10^6 qubits; both windows close when factories are available.

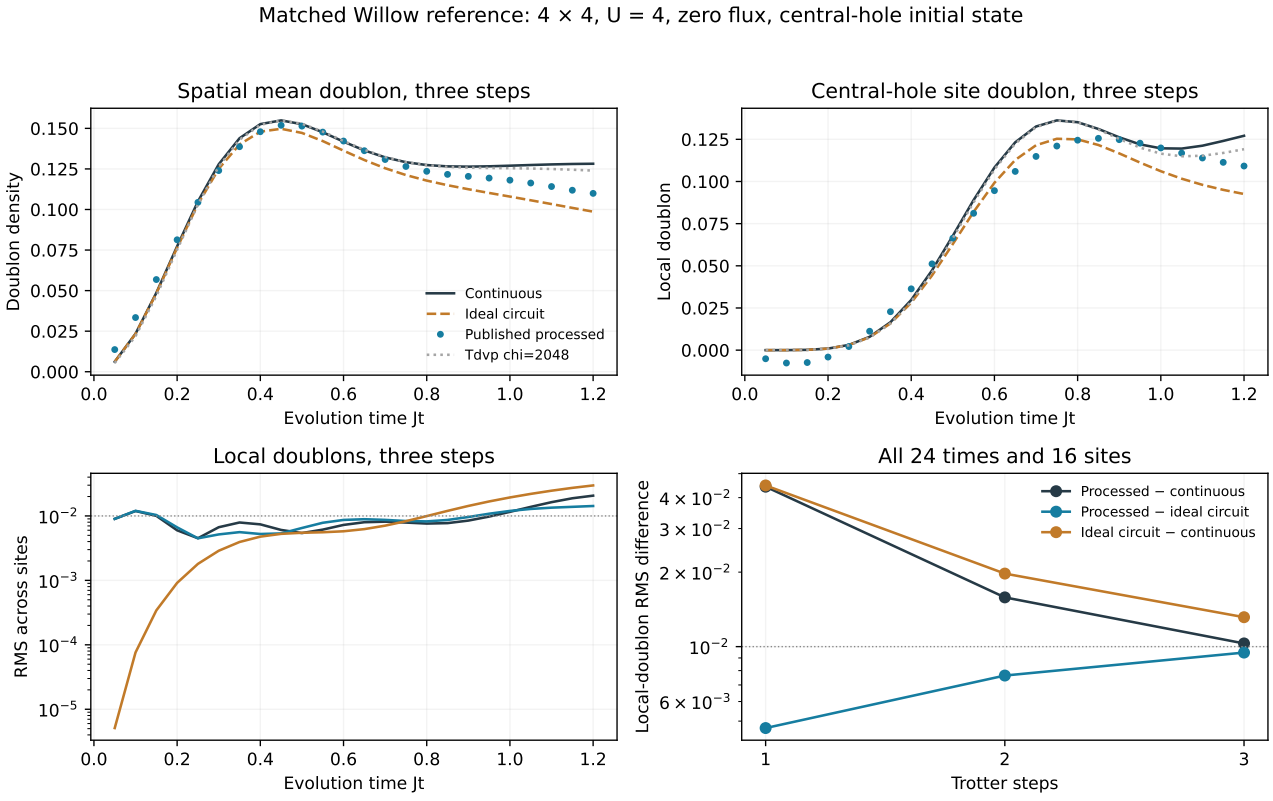
5.4 External checks

Two published operating points and one compilation check the accounting from outside. For the encoded arms, we replayed the TopQAD resource ledger for Hubbard ground-state energy estimation [18]. All 16 STAR footprints reproduce the published values, 14 of 16 runtimes without PEC agree within 2.3%, and all 16 time-optimal FT entries agree within 2.8%; the two remaining STAR rows are internally inconsistent in the publication. For physical qubits, we reconstructed the Willow 4×4, $U = 4$ circuits [14, 19] and computed ideal-circuit and continuous interacting references at all 24 saved times (Fig. 6). The processed experimental local doublon differs from continuous dynamics by 0.0103 RMS at three Trotter steps, with 71% of points within 0.01. The two contributions respond oppositely to refinement: from one to three steps the ideal circuit's error falls from 0.045 to 0.013, while the processed data's discrepancy from the ideal circuit grows from 0.0047 to 0.0095. This is the trade-off behind the physical-qubit wall. Refinement buys algorithmic accuracy with circuit exposure, and at three steps the total also benefits from partial cancellation between the two, which a controlled error budget cannot credit.

A third check tests our gate counts against compilation. CUDA-Q Logical [20] compiled the plaquette and termwise circuits at $\tau = 1$, $r = 4$, and $L = 2$–8 into Clifford+T words with GridSynth synthesis. The compiled T counts lie 0–9% below Eq. (16), and the resulting full-fault-tolerance areas agree with the formula ledger to within 0.03 in $\log_{10} Q$. The backend's physical encoded timing could not be validated, so compilation checks the counts, not the delivery model (Supplementary Material).

5.5 Which choices move the leader

The model contains choices that a simpler comparison would fix in advance: four analog lanes, one T-consumption port with Hamming-weight batches of at most 16, a data distance no smaller than the cultivation output distance, noisy PCEC for every analog rotation, and a single Trotter branch. We



Circuit references include the experiment's diagonal symmetry averaging. Experimental differences retain sampling and processing error.
The three RMS curves do not add: the signed discrepancies can reinforce or cancel. No 6 × 6 extrapolation.

Figure 6: Willow 4×4, $U = 4$ data against matched references. Processed experimental estimates are compared with ideal circuits of the same step count and with continuous interacting dynamics; the decomposition separates circuit (Trotter) error from observed experimental discrepancy. Details are in the Supplementary Material.

tested each alternative alone and all together against this reference menu, with cultivated T states throughout, which reproduces our earlier integrated model in 335 of 336 replayed cases to 10^{-9} in $\log_{10} Q$; the remaining case, a STAR design at $10^{78.7}$ qubits, lies beyond an exposure cutoff that cannot affect capacity below 10^9 qubits. Figure 7 and Table 3 attribute the changes.

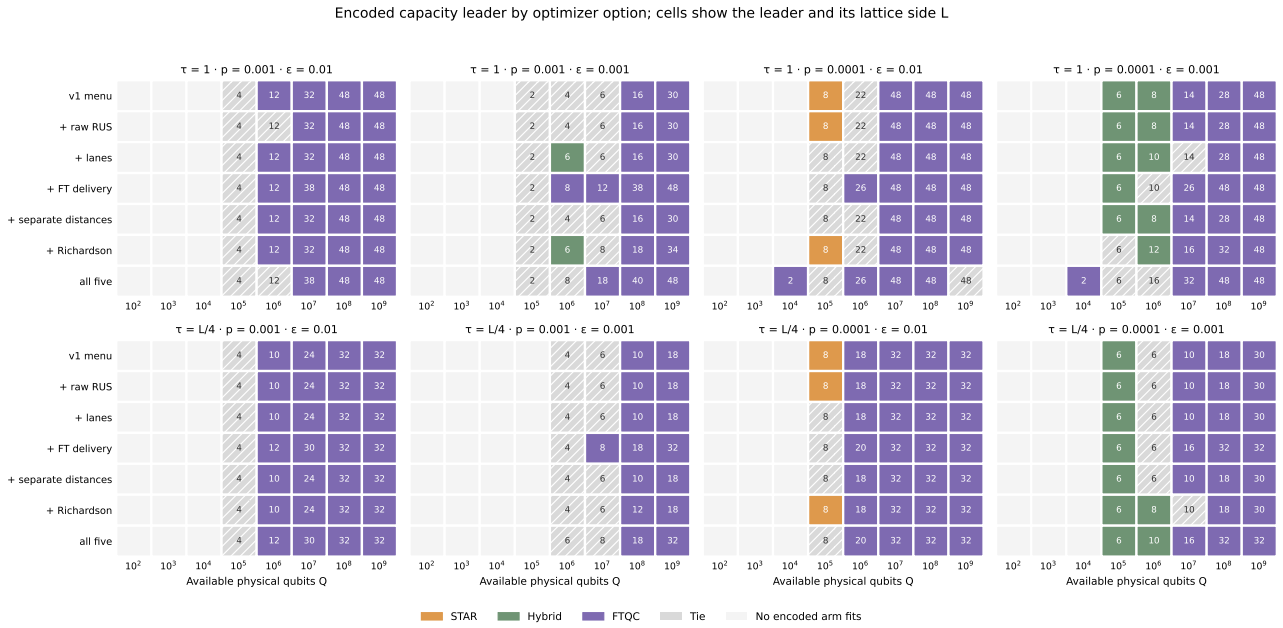


Figure 7: Encoded capacity leader for each configuration (rows), qubit budget (columns), time family, physical error, and precision, under uniform noise with the calibrated surface-code law and cultivated T states. Numbers give the leader's lattice side; hatched cells are ties. The reference menu is the top row of each panel.

Option	Cells	Mechanism
FT delivery	10	Larger Hamming-weight batches cut synthesized rotations; ports and factories cut the attempt time
Richardson	4	Base step count 8τ with a K_4/r^4 residual frees the Trotter allowance, all at $\epsilon = 0.001$
Separate distances	2	FT and hybrid run below $d = 15$ at $p = 10^{-4}$
Lanes	4	Parallel analog injection, mainly for the hybrid
Raw RUS	2	Lower analog exposure within the restricted instrument
All five	15	Mostly earlier separation of full fault tolerance

Table 3: Cells, out of 64, in which each option alters the encoded capacity leader relative to the reference menu.

Joint T delivery moves the leader most often. At 48×48, $\tau = 1$, and $\epsilon = 0.001$, whole-group phasing reduces the T count from 1.45×10^7 to 3.0×10^6, while 32 ports and 1024 factories reduce the attempt time from 149 s to 2.2 s; the required area falls from $10^{9.7}$ to $10^{8.2}$ qubits. Separate source and data distances, or optimized lanes, remove the only cells in which STAR led alone ($p = 10^{-4}$, $\epsilon = 0.01$, $Q = 10^5$, in both families). Raw RUS rarely moves the leader but supplies most of the analog gain; 16×16 STAR at $\tau = 1$ and $\epsilon = 0.01$ falls from $10^{11.7}$ to $10^{9.7}$ qubits with it alone. That gain rests on a coherent cancellation that holds when a failed trial conjugates the resource error, as teleportation does, and fails for biased branches or correlated faults (Appendix D); the conclusions about full fault

tolerance do not depend on it.

5.6 Sensitivity to the surface-code law

Replacing the reference law ($A = 0.1$, $\Lambda = 10$) by the calibrated one raises the code distance of full fault tolerance by six to eight at $p = 10^{-3}$, for example from 29 to 37 at $L = 32$, and by two to four at $p = 10^{-4}$, and delays each transition by up to a factor of two (Table 4). Because the arms protect different volumes, a shared law need not move them equally: it changes the encoded leader in 9 of 64 cells, five at fixed $\tau = 1$ and four on the time-growing workload, all at 10^5 or 10^6 qubits. At strict precision, $p = 10^{-3}$, and 10^6 qubits, for example, full fault tolerance leads alone under the reference law, while the calibrated law leaves it tied with the hybrid.

p	ϵ	Encoded beats physical	FT at $\tau = 8$
10^{-3}	0.01	$1.9 \to 3.2 \times 10^5$	$0.70 \to 1.1 \times 10^7$
10^{-3}	0.001	$2.2 \to 4.1 \times 10^5$	$2.4 \to 4.4 \times 10^8$
10^{-4}	0.01	$1.2 \to 2.0 \times 10^5$	$1.8 \to 2.6 \times 10^6$
10^{-4}	0.001	$3.8 \to 5.0 \times 10^5$	$4.9 \to 8.7 \times 10^7$

Table 4: Physical qubits at two transitions under the reference and the uniform-noise surface-code calibration, with all optimizer options and cultivated T states.

5.7 Sensitivity to delivery

Delivery is the declared assumption most likely to reorder STAR and full fault tolerance, so we varied each delivery charge around the optimized design. On the Hubbard task, allowing up to N concurrent analog lanes instead of L, as a local fermionic encoding would, changes no crossover; it only raises STAR's ceiling at $p = 10^{-4}$ and $\epsilon = 0.001$ from $L = 12$ to 14. Halving the six tiles per lane moves the edges of the analog window by 5–10%, and even the most favorable combination for STAR, N lanes of three tiles, leaves every crossover in place. Doubling them weakens the analog arms enough that full fault tolerance is never behind up to 2.6 times earlier. Quadrupling the tiles per extra T port delays full fault tolerance at strict precision from 4.7 to 5.4×10^5 qubits. The largest lever is the T source. Restricting full fault tolerance to cultivated states changes little at $\epsilon = 0.01$ and $p = 10^{-3}$, where cultivation is optimal, but strict precision needs distillation. With cultivation alone, full fault tolerance at $p = 10^{-4}$ and $\epsilon = 0.001$ is never behind only from 4.3×10^6 qubits instead of 1.9×10^5, its reach at 10^6 qubits falls from $L = 16$ to 8, and the hybrid takes the lead there. At $p = 10^{-3}$ and $\epsilon = 0.001$ it is strictly ahead only from 1.6×10^7 qubits instead of 10^6, and the $\tau = 8$ ceiling costs 4.4×10^8 qubits instead of 8×10^7. The factories chosen are two-level 15-to-1 protocols at $p = 10^{-3}$ and mostly single-level ones at $p = 10^{-4}$. The hybrid gains less from factories, because its exposure rather than its magic budget sets its size. The published factory designs, used as they stand, give every crossover to within 10%, and a declared cultivated source one hundred times better moves the strict-precision crossover at $p = 10^{-3}$ further, to 1.7×10^5 qubits. Apart from the source, the lane choices change the leader in one or two cells each.

The same runs attribute the hybrid's advantage. When the plaquette circuit's fixed $\pi/8$ rotations are injected by the STAR instrument rather than distilled, the circuit needs $10^{18.8}$ qubits at $L = 12$ ($p = 10^{-3}$, $\epsilon = 0.01$) against $10^{12.6}$ for STAR's termwise circuit and $10^{7.4}$ for the hybrid, and the hybrid's reach falls by two to six sites at every budget from 10^6 qubits. The plaquette circuit helps only once its large fixed angles leave the analog channel, exactly as the per-radian law predicts.

Two further runs bound the least certain inputs. Raising the cultivated-state error from 2×10^{-9} to 4×10^{-9}, the exact pre-growth value plus a margin for the growth stage [17], makes full fault tolerance

at $L = 6$ and 8 switch from cultivation to factories at 1.4–1.6 times the area. It moves the strict-precision crossover at $p = 10^{-3}$ from 4.7×10^5 to 7.7×10^5 qubits and changes little else. Removing raw repeat-until-success bounds the case in which its coherent cancellation fails, for example through hook errors that turn into coherent errors [21]. STAR then loses two to four lattice sites of reach and the hybrid up to two, and full fault tolerance is never behind earlier at $p = 10^{-3}$, from 1.9×10^5 rather than 3.5×10^5 qubits at $\epsilon = 0.01$ and from 1.8×10^5 rather than 4.7×10^5 at $\epsilon = 0.001$; at $p = 10^{-4}$ nothing moves (dotted curves in Fig. 2).

On random circuits, doubling the lattice-surgery time of a CNOT layer, or adding a congestion penalty growing as $\sqrt{w/16}$, changes no capability at two significant figures, because rotation layers take about $25d$ cycles against $2d$ for a CNOT layer. Lane charges shift only the edges of STAR's window, and port charges cost full fault tolerance 30% at 10^6 qubits. STAR's limit is its exposure, Eq. (20), not its delivery, so a compiled injection schedule can only add costs that the aggregate model omits.

6 Discussion

The exposure variables explain the ordering. With local full-lattice compilation, the number of rotations is $R = O(Nr)$ and their total angle is $\Theta = O(N\tau)$. Physical PEC therefore has exponent $O(pNr)$, and STAR's small-angle exposure $O(pN\tau)$, which the time-growing workload makes $O(pL^3)$; injection floors, growth, and the fixed plaquette angles restore some growth with r. Full fault tolerance instead needs $O(Nr \log(Nr/\epsilon))$ T states and a code distance logarithmic in the protected volume, at the price of a large starting area. Every arm has the logarithmic sampling allowance of Eq. (5). An exponent that grows as a power of the lattice therefore meets a wall, and only full fault tolerance, whose sampling exponent is zero, keeps scaling. The two workloads isolate the per-radian term. With rotations of order one, STAR falls three orders of magnitude behind full fault tolerance at 10^6 qubits; with small Trotter angles it stays within a few lattice sites until a few times 10^5 qubits. The controlled sweep of Sec. 5.2 removes the other differences between the workloads and recovers Eq. (20) directly.

Several limitations remain. Lane, port, and transfer charges are declared rather than compiled, and lattice surgery is calibrated only for isolated adjacent gadgets, under the earlier noise model with one-qubit error $p/10$. The interacting Trotter coefficient is converged near $\tau = 1$ on patches up to 8×2, 6×3, and 4×4; beyond that it is a transfer bounded by every measured patch, and it is extrapolated beyond $\tau = 4$. The surface-code law is calibrated for memory at $d \leq 9$ and extrapolated to the distances used; a single law is statistically rejected, and at $p = 10^{-3}$ its slope exceeds the largest-distance slope by about 5%, which could cost roughly one code distance at $d \approx 25$. The factory costs replace the error law inside a published model rather than simulate factories. The effective physical-qubit mitigation overhead is assumed, and physical estimators with controlled bias at our absolute accuracy have not been demonstrated at these sizes. The raw-RUS cancellation assumes unbiased, independent retries and has not been checked in a circuit-level simulation of injection; the PCEC-only curves bound its failure.

The most useful next calculations are a circuit-level simulation of analog injection, which would test the raw-RUS cancellation that supplies most of STAR's gain; interacting Trotter coefficients at $\tau \geq 2$ on wider lattices; and simultaneous lattice surgery for many long-range CNOTs. The first two could move STAR's plateau. None is likely to move the crossovers of full fault tolerance, which rest on its T source, and a compiled schedule for parallel analog injection can only make STAR's costs higher.

7 Conclusions

We compared error mitigation, STAR partial protection, a hybrid, and full surface-code fault tolerance on one nearest-neighbor array, for random QUOPS circuits and for Fermi–Hubbard dynamics, organizing the comparison by the largest computation that fits a qubit budget within one week. A common accounting model optimized each architecture's step count, error allocation, code distance, and delivery, with a circuit-level calibration of the surface-code law under uniform noise. At $p = 10^{-3}$ on random circuits, physical qubits score about 240, STAR saturates near 6×10^3, and full fault tolerance scores 4.8×10^6 with 10^6 qubits. On the Hubbard task, physical-qubit computation saturates at $\tau \approx 1.5$, STAR and the hybrid at $\tau \approx 2.5$–3, and full fault tolerance is never behind from about 3.5×10^5 physical qubits at $\epsilon = 0.01$ and 4.7×10^5 at $\epsilon = 0.001$, reaching the largest simulated evolution with 10^7 qubits at $\epsilon = 0.01$. At strict precision these crossovers rest on distillation factories, which we priced with the same calibrated surface-code law. A controlled angle sweep confirms that STAR's reach scales as the inverse of its mean injected angle, and the delivery sensitivity shows that STAR is limited by that exposure rather than by how fast it injects. The hybrid's advantage over STAR comes from distilling its fixed $\pi/8$ rotations.

A Trotter step law and finite-patch checks

This appendix gives the exact dynamics behind Eq. (10): direct step checks on 4×3 and 4×4 patches with their time dependence, the matrix-free reference that computes them, exact strips up to 8×2 and 6×3, and the free-fermion limit to 24×24. Earlier small-patch calibrations and a conservative truncation bound are in the Supplementary Material.

A.1 Direct step checks and finite-patch time dependence

We compute the exact reference evolution in the $N_\uparrow = N_\downarrow = 8$ sector of the 4×4 OBC lattice. Its $\binom{16}{8}^2 = 165{,}636{,}900$ amplitudes fit a matrix-free calculation with four complex state vectors, using about 10 GB of memory. At $\tau = 1$, the central doublon at coordinate $(2, 2)$ is 0.1452313819. The sampling convention remains $D_* = 0.1$ and the absolute targets remain 0.001 and 0.01; we do not silently redefine the requested precision using the newly calculated signal.

The plaquette coefficient $|D_r - D_{\text{exact}}|r^2$ changes from 0.262431 at $r = 16$ to 0.261828 at $r = 32$, a 0.23% difference. It is about 20% above the 3×3 value. The termwise coefficient changes from 0.188386 to 0.187732, a 0.35% difference, and is about 2.45 times its 3×3 value. The 4×3 coefficients, 0.2453 and 0.1392, lie between the square-patch results, as Figure 8 shows. The earlier 10% planning margin therefore does not cover even this $\tau = 1$ transfer. Exact finite-lattice evolution resolves that specific uncertainty; it does not establish an infinite-lattice coefficient.

Validate before reallocating. The initial 4×4 resource sensitivity interpolated between measured steps using C/r^2, with a 10% margin. We now directly evaluate every selected step count for the headline reference designs: effective and two PEC physical-qubit models, reference STAR and hybrid, and reference, single-fast-block, and four-fast-port FT. The two precision targets give 16 allocations sharing nine circuits. All satisfy

$$|D_r - D_{\text{exact}}| + 10^{-10} \le \epsilon f_T^{\text{saved}}, \tag{21}$$

using their *original* Trotter fraction f_T^{saved}. The measured errors use 89.3%–90.9% of these allowances. No change to statistics or another error component is needed to pass these checks.

We then separately reoptimize with the measured Trotter error and the same 10^{-10} numerical allowance. Table 5 shows the strict estimates at each originally selected r. Reference FT falls from 1.88 to 1.66 million qubits, and the single fast block from 1.16 to 0.99 million, because the smaller charged Trotter error leaves more allowance for sampling. STAR and hybrid remain single-copy calculations. The saved Trotter allowances were already adequate; these reductions come from using their remaining allowance. Searching all measured steps gives the same headline areas, sometimes at a different r.

Checking the selected physical-qubit step count $r = 21$ removes the earlier artificial restriction to $r = 8, 16, 32$, which had inflated its measured-menu cost to 2196 qubits. This validates the Trotter component only: plaquette accuracy is still transferred to the experimental gate-count proxy, and mitigation bias remains unmeasured. At loose precision the two-qubit PEC scenario also fits one 36-qubit copy at $r = 8$, whereas its strict estimate exceeds 10^{12} qubits. The full-noise PEC proxy is much more costly. These differences remain visible in the saved tables and should not be read as an architecture-independent physical-qubit limit.

Longer time and the capacity sensitivity. We extend the same exact 4×4 reference and both product formulas to $\tau = 2, 4$. At each time, the initial counts are $8\tau, 16\tau, 32\tau$; further doubling is required if the two finest coefficients disagree by more than 5%. Figure 9 compares the converged tails

Design	Saved r	Saved Q	Measured-error budget Q
Physical, effective	21	72	72
STAR reference	33	71,442	71,442
Hybrid reference	41	73,348	73,348
FT reference	33	1,875,882	1,655,190
FT single fast block	29	1,155,490	988,112
FT four fast ports	33	385,052	385,052

Table 5: Direct 4×4 checks at $\tau = 1$, $\epsilon = 0.001$. Both columns use the same step count, hardware, and source menu. The last column reallocates the smaller measured Trotter error before minimizing resources. The effective physical model retains its assumed mitigation bias and variance; four fast ports retain the uncompiled throughput assumption.

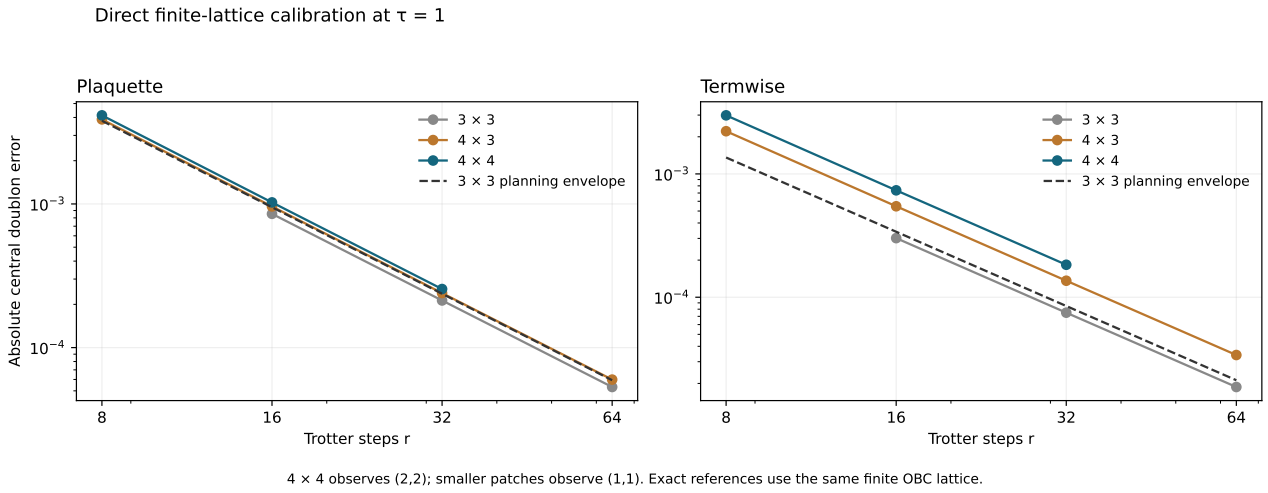


Figure 8: Direct Trotter errors at $\tau = 1$ on three finite patches, with exact references on each same lattice. Both orderings exhibit r^{-2} convergence, with coefficients that depend on the patch. The dashed line is the previous 3×3 planning envelope, including its 10% margin. The 4×4 observable is at $(2, 2)$; smaller patches use $(1, 1)$. This comparison calibrates the finite 4×4 example and does not certify large-L or long-time transfer.

with the smaller patches. The observed site remains $(2, 2)$ and each product formula is compared with exact evolution on that same lattice. The converged-pair coefficients on 4×4 are 1.096 and 0.775 for plaquette and termwise evolution at $\tau = 2$, and 4.193 and 3.029 at $\tau = 4$. The finest pairs agree within 0.36% in all four cases. The $\tau = 4$ plaquette value is close to 4×3 and below 3×3; the termwise value exceeds both smaller patches. A single multiplicative finite-size correction would miss this dependence on ordering and time.

A.2 Matrix-free finite-lattice reference

Writing the state as a matrix indexed by up- and down-spin occupation masks, we apply

$$H\Psi = h_{\uparrow}\Psi + \Psi h_{\downarrow}^{\mathsf{T}} + V \odot \Psi. \tag{22}$$

Each hopping matrix is sparse within a single spin sector; no full-sector Hamiltonian is stored. Fermionic signs count occupied sites between the hopping endpoints in the snake order. For $U/J = 4$,

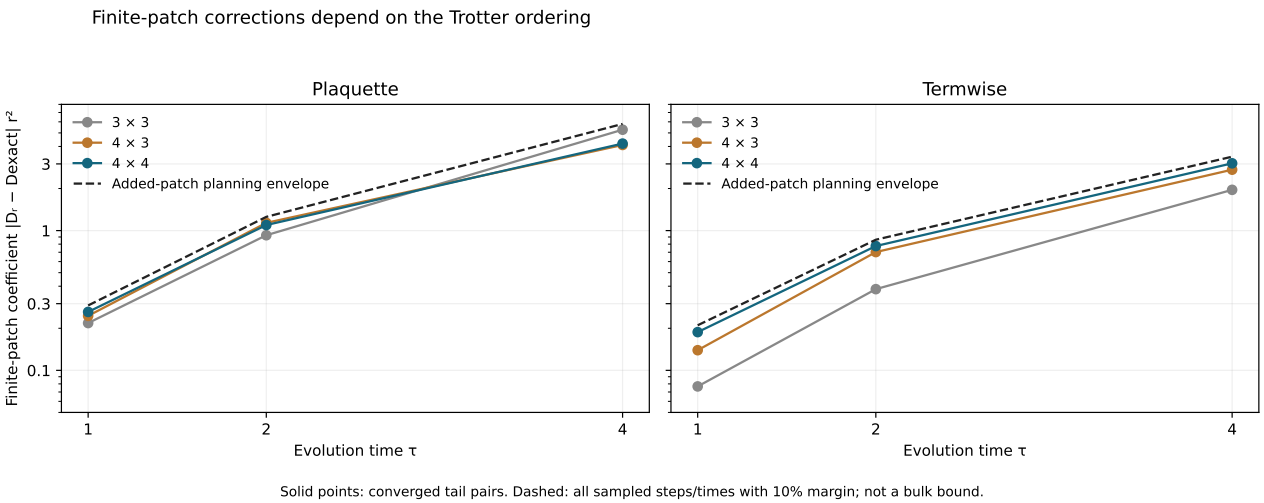


Figure 9: Ordering-specific Trotter coefficients through $\tau = 4$ on three finite patches. Solid points use converged tail pairs; the dashed envelope also covers measured coarser-step coefficients and earlier sampled times, with a 10% margin. The larger patches have only three time nodes; unsampled time peaks and larger-lattice corrections remain possible.

the triangle bound is $\|H\| \le 2|E| + N = 64$ on 4×4. We evaluate

$$e^{-iH\tau}\psi_0 = J_0(a)\psi_0 + 2\sum_{k\ge 1}(-i)^k J_k(a) T_k(H/64)\psi_0, \qquad a = 64\tau, \tag{23}$$

using the three-term Chebyshev recurrence. The bound

$$|J_k(a)| \le \frac{(|a|/2)^k}{k!}\exp\left(\frac{a^2}{4(k+1)}\right) \tag{24}$$

follows by bounding the absolute Bessel series; successive right-hand sides have ratio at most $|a|/[2(k+1)]$. A geometric tail therefore selects a degree with state truncation error below 10^{-12}. For a $[0,1]$ observable, state error η gives observable error at most $2\eta + \eta^2$. This controls polynomial truncation, separately from floating-point error. The computed 4×4 norm error is about 10^{-13}.

Trotter hopping stages use exact two-state rotations. Plaquette stages use the fermionic Hadamard decomposition already employed by the compiler. Complete state vectors for the exact reference and both orderings agree with the independent Python implementation on 2×2, 3×2, and 3×3 to better than 2×10^{-11}. Large-run outputs retain all-site doublons, norms, input and executable hashes, and atomic case records. The step convergence criterion is a 5% agreement between the two finest observable-error coefficients, not a proof of bulk convergence.

A.3 Exact strips beyond 4×4

A fermionic matrix-product state that applies the same termwise gates reproduces the sector solver exactly on small patches but cannot reach larger ones: the quench from the Néel state builds near-volume-law entanglement by $\tau = 1$, and on 4×3 a bond dimension of 256 already discards 43% of the weight. We therefore raised the sector solver's limit to 18 sites and computed every strip of width 2 up to 8×2 and of width 3 up to 6×3 (2.4×10^9 amplitudes), with exact references at $\tau = 1$ and 2 (Fig. 10).

At $\tau = 1$ the termwise coefficient rises to 0.19 at length 5 and stays within 0.18–0.19 to length 8, and widths 2, 3, and 4 give 0.186, 0.191, and 0.188; the growth from 3×3 to 4×4 was a short-patch

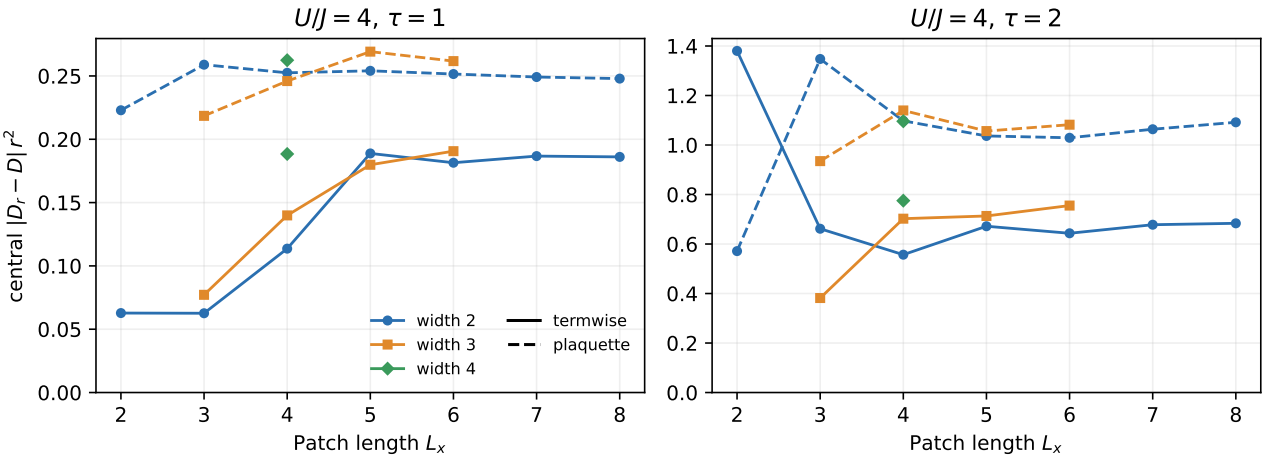


Figure 10: Exact central-doublon Trotter coefficients at $U/J = 4$ versus patch length, for widths 2 (circles), 3 (squares), and 4 (diamond), termwise (solid) and plaquette (dashed); $r = 16$ at $\tau = 1$ and 32 at $\tau = 2$.

effect. The plaquette coefficient is flat at 0.25–0.27. At $\tau = 2$ both converge in length by length 5–7, but the termwise value rises slowly with width, 0.68, 0.76, and 0.78 for widths 2, 3, and 4. Every measured value lies below the planning coefficients of Eq. (10), 0.21 and 0.29 at $\tau = 1$ and 0.86 and 1.25 at $\tau = 2$, so no optimized step count changes.

A.4 Free-fermion size dependence

At $U = 0$ every gate of both circuits, including the fermionic Hadamards and the Jordan–Wigner parity strings, is a rotation generated by a Majorana bilinear. The circuits are therefore Gaussian, and we evolve the $4N \times 4N$ Majorana covariance matrix exactly, one gate at a time, from the Néel state; the central doublon follows from Wick's theorem. The gate sequences are those of the compiler, and the Gaussian doublons agree with state vectors on 2×2 and 3×3 to 10^{-14}. Figure 11 shows the coefficient at $r = 32$, which agrees with $r = 16$ to a few percent.

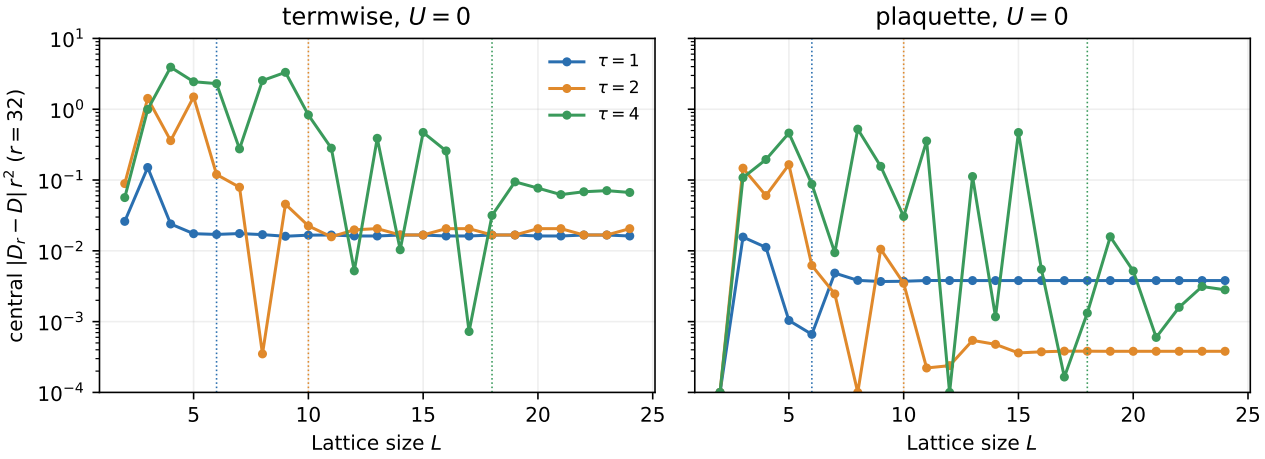


Figure 11: Central doublon Trotter coefficient at $U = 0$ versus lattice size, at $r = 32$. Dotted lines mark $L = 4\tau + 2$.

At fixed τ the coefficient becomes independent of L once the lattice is wider than the light cone, $L \gtrsim 4\tau + 2$. Beyond that size the Jordan–Wigner ordering of the termwise circuit adds no size dependence. Inside the light cone, boundary reflections make small patches fluctuate by up to two orders of magnitude, mostly upward: at $\tau = 4$ the termwise 4×4 value is 3.9 against a bulk value of 0.07. On the time-growing family $\tau = L/4$ the central coefficient stays below 0.5 up to $L = 24$, whereas the interacting planning coefficients of Eq. (10) are 0.21 at $L = 4$ and 13 at $L = 24$. The interaction commutators that dominate at $U/J = 4$ are absent here, so the free values are 10–100 times smaller and do not replace the calibration. What carries over is the structure: in the free limit the coefficient is set by the light cone, and finite patches err on the high side.

B Circuit-level calibration of the surface-code law

We calibrate Eq. (13) with a rotated-surface-code memory experiment under standard uniform circuit noise: Stim's rotated memory circuit with native CNOTs, depolarizing error p after every one- and two-qubit gate and on every idle qubit in every layer (including the measurement and reset layers), and flips of probability p after resets and before measurements. This is the convention of the cultivation and distillation estimates we use. Resource timing keeps the declared 240 ns round. Circuits are sampled with Stim and decoded by minimum-weight perfect matching in PyMatching [15, 16].

The fit uses 7.8×10^7 training shots at $R = 10d$ and $20d$ rounds and 1.8×10^7 independent holdout shots at $R = 100d$, for $d = 3, 5, 7, 9$ at $p = 10^{-3}$ and $d = 3, 5, 7$ at $p = 10^{-4}$. Each basis $b \in \{X, Z\}$ has its own amplitude and a common suppression factor,

$$q_b(d) = A_b \Lambda^{-(d+1)/2}, \qquad P_b(R) = \tfrac{1}{2}\left[1 - (1 - 2q_b)^{R+\delta_{db}}\right], \tag{25}$$

where $P_b(R)$ is the probability of a logical failure after R rounds and δ_{db} absorbs preparation and readout at the endpoints. We use the conservative sum $A = A_X + A_Z$ as the per-patch amplitude.

p	Λ [95% interval]	A	Note
10^{-3}	6.30 [6.18, 6.43]	0.103	uniform noise, $d = 3$–9
10^{-4}	44 [41, 46]	0.186	uniform noise, $d = 3$–7
10^{-3}	7.41 [7.20, 7.62]	0.077	earlier model, one-qubit error $p/10$
10^{-4}	89 [64, 146]	0.125	earlier model, one-qubit error $p/10$

Table 6: Fitted surface-code law. Intervals are asymptotic profile-likelihood intervals conditional on Eq. (25). The earlier model used a CZ schedule with one-qubit error $p/10$ and is retained for comparison.

A single law is strongly rejected (deviance 680 on 5 degrees of freedom at $p = 10^{-3}$ and 341 on 3 at $p = 10^{-4}$), because with this many shots the distance-to-distance slope is resolved: at $p = 10^{-3}$ it is 5.6, 7.0, and 6.0 for $d = 3 \to 5 \to 7 \to 9$, so the fitted $\Lambda = 6.30$ exceeds the largest-distance slope by about 5%, which could cost roughly one code distance at $d \approx 25$. At $p = 10^{-4}$ the slopes are 41 and about 110, so the fit is conservative there. Refitting with $d \geq 5$ only, with no endpoint correction, or on the holdout changes the required distance by at most two. Eq. (13) is therefore an approximate resource model rather than an exact description. Relative to the reference $A = 0.1$, $\Lambda = 10$, the fitted law raises the distance needed for a protected volume of 10^8 patch-rounds and budgets 10^{-4}–10^{-6} from 21–25 to 27–33 at $p = 10^{-3}$, a factor 1.5–1.8 in patch area. Every such distance lies beyond the sampled range, and correlated or coherent hardware noise is not included. The evidence is in `results/surface_law_sd6_v1`; the earlier calibration is in `results/surface_law_v1`.

Lattice surgery. With the earlier noise model (one-qubit error $p/10$) we simulated isolated ZZ and XX merges, with seams of one to $3d$ columns, and a full CNOT from an L-shaped control, ancilla, and target, at $d = 3$–9 and both error rates: 678 million shots in 158 cells, with ideally encoded inputs. With $R_m = d$ merge rounds, every resolved cell fails at 0.73–1.15 times $Vp_L(d)$, where V is the gadget's space-time volume in patch-rounds, so the memory law needs no surgery penalty at the factor-of-two level; $d = 7$ at $p = 10^{-4}$ is unresolved. The CNOT takes $2d \times 240$ ns plus 300 ns, so our two clocks per layer omit 9–21% of its time at $d = 7$–3, which the delivery sensitivity shows is immaterial. Shorter merges add a time-like failure term that has no universal slope and exceeds the memory law by up to 57 times, so merges are kept at d rounds. Simultaneous routing of many long-range CNOTs, and therefore congestion, is not covered, and the calibration has not been repeated under uniform noise; its conclusion is a ratio to the memory law under the same noise and is expected to carry over. Evidence is in `results/surgery_v1`.

Distillation factories. The factory model of Ref. [8] prices every patch, merge, and storage step with one per-patch, per-cycle logical error, $0.1(100p)^{(d+1)/2}$, which is the reference law above. We ran the paper's public code with Eq. (13) in its place, keeping its injection model and protocol structure, and first confirmed that with the reference law it reproduces all eleven published designs we use. We then re-optimized the distances of one-level 15-to-1, two-level (15-to-1)×(15-to-1), and (15-to-1)×(20-to-4) factories over 9255 designs at 40-digit precision, and kept, per decade of output error, the designs with the smallest footprint and the smallest space-time volume per output. At $p = 10^{-3}$ the published designs' output errors grow 30 to 4×10^4 times under the calibrated law, for example from 3.3×10^{-14} to 1.7×10^{-10}; re-optimized two-level designs reach the published output errors with about 50% more qubits, and one-level designs must be replaced by two-level ones. At $p = 10^{-4}$, where the fitted law is conservative, the re-optimized designs need 90–350% more qubits. This replaces the error law inside the published model; it is not a circuit-level simulation of a factory. Evidence is in `results/factory_calibration_sd6_v1`.

C STAR channels, growth, and retries

The STAR instrument of Sec. 4.3 evaluates the transversal multi-rotation preparation exactly at every workload angle and composes it with measured growth. This appendix gives the retry structure, the accepted-channel checks, and the growth campaigns behind that model.

C.1 Explicit retries, covariance, and parallelism

The logical injection instrument prepares $|A_\theta\rangle = e^{-i\theta Z}|+\rangle$, applies $\mathrm{CNOT}_{\mathrm{data}\to\mathrm{ancilla}}$, and measures the ancilla in Z. Its data Kraus operators are

$$K_0 = 2^{-1/2}e^{-i\theta Z}, \qquad K_1 = 2^{-1/2}e^{+i\theta Z}. \tag{26}$$

After outcome one, an injection of angle 2θ corrects a successful next branch: $K_0(2\theta)K_1(\theta) = K_0(\theta)/\sqrt{2}$. Repeating this gives the standard angle-doubling protocol underlying the STAR reference [2, 3]. CUDA-Q P0 now contains explicit preparation, CNOT, measurement, conditional retries, and a returned exhaustion flag for caps $1, 2, 4, 8, 24$. This compilation does not supply a surface-code injection or patch-growth implementation.

For a Pauli fault on either wire immediately before or after CNOT, each conditional K remains diagonal or antidiagonal with equal-magnitude entries. Hence

$$K^\dagger K = I/2, \qquad KZ = \pm ZK. \tag{27}$$

The first identity makes the outcome probability independent of the data state; the second makes the conditional map commute with the channel $\mathcal{Z}(\rho) = Z\rho Z$. It therefore commutes with a Z-dephasing inverse, which is a linear combination of $\mathcal{I}$ and $\mathcal{Z}$. Mixtures of such faults and compositions over retries retain this local covariance under the stated branch-independent Pauli model. All sixteen two-qubit Paulis at both locations are checked independently.

This verifies the needed local commutation condition for that declared instrument. It does not establish branch-independent residual noise for physical state preparation, patch growth, measurement-dependent selection, or a mismatched channel inversion. The worst-case Γq_L bias bound therefore remains the default; q_L is a separately labeled covariance-based reference scenario. The physical STAR schedule still has to justify using it.

For b simultaneous ideal injections with geometric success probability $1/2$ and cap K, the mean batch retry time in units of one attempt is

$$\mathbb{E}[M_b] = \sum_{k=0}^{K-1} \Pr(M_b > k) = \sum_{k=0}^{K-1} \left[1 - (1 - 2^{-k})^b\right]. \tag{28}$$

It approaches 2 for one lane, but is approximately 3.50 for four lanes. Simply dividing the serial duration by four misses the slowest retry. The implemented policy has a barrier between batches, reserves six injection tiles per lane, and assumes nine clocks per attempt, recovering the previous 18-clock serial mean. It forms disjoint batches using full JW supports, not merely disjoint fermionic endpoints. An asynchronous policy could improve this conservative batch schedule.

With R rotations the ideal no-exhaustion probability is $(1 - 2^{-K})^R$; the cap K is chosen from a small allocated tail allowance, and the acceptance penalty is charged.

C.2 STAR: accepted channels and growth components

We enumerate the projected error-angle mixture for the multirotation construction [3]. For odd injection distance d_i, let $k = (d_i + 1)/2$ and $\alpha = \arctan(|\tan\theta|^{1/k})$. With ideal syndrome projection, the no-fault acceptance is

$$P_{\text{acc}}^{(0)} = \cos^{2k}\alpha + \sin^{2k}\alpha. \tag{29}$$

Independent rotation-generator Pauli faults have probability $p/15$ on weight-two factors and $p/30$ on the final weight-one factor. Enumerating the surviving branches gives weights w_j and angle errors δ_j, with coherence multiplier $c = \sum_j w_j e^{-2i\delta_j}$. This is an exact calculation for these faults and perfect projection; the other physical faults of an injection circuit are not included. Among them are hook errors, which in T-state cultivation propagate to coherent errors after the final check and lower the fault distance [21]. A coherent error of this kind need not be conjugated on a failed teleportation, so it would weaken the raw-RUS cancellation of Appendix D; the PCEC-only curves of Fig. 2 bound that case.

Ideal probabilistic coherent-error cancellation (PCEC) composes this channel with the conjugate mixture, giving $|c|^2$ and hence a Pauli channel. We also execute one level of noisy PCEC in the model: sampled inverse-angle corrections themselves use raw repeat-until-success (RUS) channels. Their angle-dependent preparation time, retries, and growth noise are charged. The resulting imaginary component is retained, rather than discarded by an assumed twirl. For a histogram with rotation multiplicities R_j, the separate ledger is

$$\chi = -2\sum_j R_j \log(\operatorname{Re} c_j), \qquad B_{\text{coh}} \le \tfrac{1}{2}\left[\prod_j \left(1 + \frac{|\operatorname{Im} c_j|}{\operatorname{Re} c_j}\right)^{R_j} - 1\right]. \tag{30}$$

Here χ is the dephasing-component inverse-norm estimate, while B_{coh} bounds the coherent residue after that inverse; the product accounts for its non-CPTP norm. It is not a claim that the complete noisy gadget is Pauli. Workload retry caps reserve 1% of ϵ for omitted branches; the actual aggregate tail is recorded.

Detector rejection and high-distance growth remain explicit scenarios. We use $P_{\mathrm{acc}} = P_{\mathrm{proj}}(1 - c_d p)^{d_i^2+6d_i}$, with $c_d = 1, 4$, and independent growth dephasing

$$f(d_i, p) = \min\left\{0.25, f_9 \left(\frac{p}{10^{-3}}\right)^5 \left(\frac{d_i}{9}\right)^2 \left(\frac{0.01}{p}\right)^{-(d_i-9)/2}\right\}, \quad f_9 \in \{0, 10^{-5}, 10^{-4}\}. \tag{31}$$

There is no growth term when $d_i = d_{\mathrm{data}} = 25$. The f_9 values are anchored at $p = 10^{-3}$; their assumed extension scales as $p^{(d_i+1)/2}$ at fixed distance. These coefficients are not transplanted calibrations from Chung et al., whose reported noise model differs [18]. Switching chooses the original-injection error reference when it is smaller after growth; its error and high-acceptance service remain declared assumptions. The study therefore tests a mechanism and its sensitivities, not the feasibility of a complete optimized source.

A separate circuit-level experiment tests direct rotated-patch growth from $d_i = 3, 5$ to $d_f = 5, 7$, with no-growth controls. The starting patch is ideal; fresh right and bottom strips are prepared in Z and X respectively, then the target patch is stabilized for d_f rounds. Stim supplies a nearest-neighbor extraction schedule; transition detectors are constructed from exact stabilizer flows and later detectors compare adjacent rounds. PyMatching decodes the circuit. Gate, reset, measurement, and idle errors are included, with the 240 ns cycle explicitly charged. Every one of 20 cases uses 500,000 shots, including both X/Z tests and both physical error rates. Noiseless and deliberately inserted logical-flip controls check that the detector and logical records are meaningful. These memory components include final readout and exclude noisy initial injection; they cannot establish the distance-25 arbitrary-angle growth floor.

A larger fixed-shot campaign extends the same protocol to initial distances $d_i = 5, 9, 13$ and final distances $d_f = 13, 25$ at $p = 10^{-3}, 10^{-4}$ in both bases. All 24 cases complete one million shots. The model uses the campaign repeated under uniform noise, in which the growth circuit's one-qubit and idle errors rise from $p/10$ to p and its time-based reset allowance is removed, with the schedule and detectors unchanged. At $p = 10^{-3}$, $5 \to 25$ then produces 119 X-basis and 819 Z-basis failures, against 51 and 419 under the earlier model, and $5 \to 13$ produces 108 and 407. Cases from $d_i = 9$ and 13 at $p = 10^{-3}$ see at most seven failures, and every case at $p = 10^{-4}$ sees none. A zero count has simultaneous one-sided upper bound 6.174×10^{-6} per component, using a 5% family error budget over all 24 intervals. If the two basis tests describe a common Pauli channel, their union bounds its total error by 1.235×10^{-5} for zero-failure pairs. Evidence is in `results/growth_sd6_v1`. Thus these data constrain growth at the data-code distance but do not yet certify a 10^{-5} total floor. They also show that the initial patch distance matters even when the final patch is large. Initial injection faults remain excluded, and we do not subtract the no-growth control from a measured campaign error.

D Raw repeat-until-success and PCEC

At retry m, let the accepted rotation multiply off-diagonal elements by $c_m = 1 - a_m + ib_m$ with $a_m, b_m = O(p)$. Assume independent faults between retries, success probability exactly one half, the conjugate multiplier c_m^* on failure, and ideal angle doubling. Averaging over the RUS branches then gives the channel multiplier

$$C = \sum_{n\geq 0} 2^{-n-1} c_n \prod_{m<n} c_m^* = 1 - \sum_{m\geq 0} 2^{-m} a_m + O(p^2), \tag{32}$$

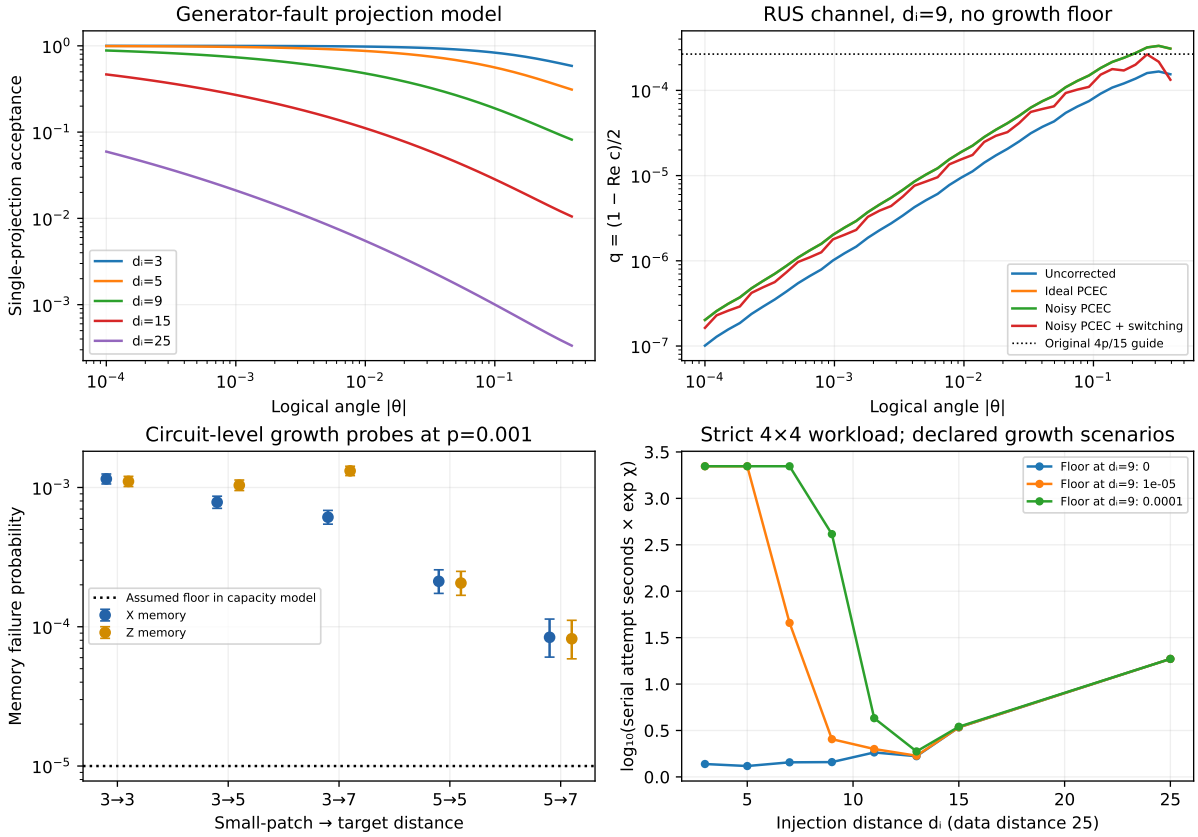


Figure 12: STAR mechanism and growth checks. Upper panels use the specified rotation-generator fault submodel; $q = (1 - \mathrm{Re}\, c)/2$ alone does not describe its coherent component. Lower left: decoded X/Z memory error for ideal small-patch preparation followed by noisy growth, stabilization, and readout, with fixed-sample 95% binomial intervals. The dotted floor is a capacity-model assumption, not a fitted growth curve. Lower right: serial service times multiplied by the dephasing sampling factor for a fixed strict-precision 4×4 workload, under declared growth scenarios. A complete injected-state error model is still required to replace the STAR capacity assumptions.

because the first-order coefficient of each b_m is $2^{-m-1} - \sum_{n>m} 2^{-n-1} = 0$. Coherence therefore cancels at first order, while dephasing retains the weights 2^{-m}. Ideal PCEC instead maps each c_m to $|c_m|^2$, which removes the coherent term of every trial but doubles its dephasing to leading order.

Numerically, over 27 angles in $[10^{-3}, \pi/8]$, injection distances 5, 9, and 25, and $p = 10^{-3}$ and 10^{-4}, the PCEC exposure exceeds the raw exposure by a factor 1.999–2.007 without growth, and $|\,\mathrm{Im}\, C_{\mathrm{raw}}|$ stays below 2.4×10^{-7}, falling a hundredfold when p falls tenfold. With the growth bound composed, the ratio falls to 1.5–1.6. The cancellation fails for a biased success probability, for same-sign success and failure errors, and for rare correlated faults; the script `python/raw_rus_check.py` quantifies each case.

Toshio *et al.* [3] average the same branches in their Appendix E, yet report a raw-RUS prefactor slightly larger than with PCEC and a logarithmic dependence on the target angle. Their model fixes everything except one sign. A resource prepared for angle $a > 0$ is wrong with probability $P_L(a) = a^{2/k} P_{\mathrm{ud}}$ and then over-rotates by $\Delta(a) = -a^{1-2/k}$, and completion at trial K composes $\mathcal{E}_{2^{K-1}\theta} \circ \mathcal{E}_{-2^{K-2}\theta} \circ \cdots \circ \mathcal{E}_{-\theta}$; the over-rotation is stated only for positive angles. A failed teleportation implements minus the resource's actual angle, so its error is $-\Delta$. With this sign, their averaged coherent term vanishes to first order ($|\tilde{y}|/\tilde{x} \approx 10^{-17}$ for $\theta = 10^{-4}$–$\pi/8$ and $k = 2$–6), and raw RUS

carries exactly half of the PCEC dephasing, as in our instrument. Evaluating Δ at the absolute angle instead makes the coherent terms add over the $\log_2(1/\theta)$ retries and reproduces both features of their Appendix E: raw RUS above PCEC at small angles, growing logarithmically (for $k = 4$, from $0.17k$ to $0.70k$ in units of θp as θ falls from 0.05 to 3.5×10^{-4}). The script `python/rus_reconciliation.py` computes both. Our asymptotic forms of P_L and Δ match their PCEC prefactor at $k = 3$ but not uniformly in k, because they evaluate exact expressions; the comparison of signs is internal to one model. Raw RUS is accordingly an option of the optimizer, charged with its coherent residue and retry tail, rather than a conclusion that PCEC is unnecessary. It changes the capacity leader in two of 64 cells, so the conclusions about full fault tolerance do not depend on it; the STAR and hybrid capacities do.

E Physical-qubit estimators

The physical arm in Fig. 2 brackets an unknown estimator between PEC of every gate and an assumed effective overhead. This appendix collects the inverse problem, which states what a physical estimator would need in order to compete, together with the compilation and noise details behind Eq. (12).

E.1 The physical circuit: Jordan–Wigner with fermionic swaps

The Jordan–Wigner encoding orders the $2N$ spin modes along a line, here the row snake of Fig. 13(a). A hop between modes k positions apart becomes a Pauli string with Z on the $k-1$ qubits between them. Horizontal neighbors are adjacent in the snake, but vertical neighbors can be up to $2L-1$ positions apart. Compiled directly on a nearest-neighbor array, each such string needs a parity ladder of $2(k-1)$ CNOTs and depth growing with k [Fig. 13(b)].

A fermionic swap exchanges two adjacent modes and applies a phase of -1 to $|11\rangle$, which keeps the Jordan–Wigner order consistent after the exchange. Layers of fermionic swaps on alternating pairs reverse the order of a line of modes, and during that reversal every pair of modes is adjacent exactly once. Each hop is applied at that moment, merged with its swap, so every gate acts on neighboring qubits and no long Z string appears [Fig. 13(c)]. On the full lattice such sweeps cost $O(L)$ layers, so a Trotter step needs $O(NL)$ two-qubit gates [13]. This is the origin of the $G \propto rL^3$ scaling in Eq. (12), which we anchor to the published Willow compilation. A local fermionic encoding removes the swaps and gives $G \propto rL^2$, but in the compact-encoding scenario of the Supplementary Material it does not reduce the gate count at the lattice sizes the physical arm can reach.

E.2 The inverse problem

This calculation compares with the reference menu at fixed $\tau = 1$ (Sec. 5.5); it states requirements on an unknown estimator and does not use the optimized encoded designs.

For a fixed physical estimator, let F be its variance per attempted circuit relative to $v_* = D_*(1 - D_*) = 0.09$, and let b be its remaining systematic offset. Rejection is included in F. At footprint a, attempt duration t_a, and deadline T, the evaluation budget after training is

$$M = \max\left\{0, \left\lfloor \frac{Q}{a} \right\rfloor \left\lfloor \frac{T}{t_a} \right\rfloor - M_{\text{train}}\right\}. \tag{33}$$

The Gaussian planning condition and its two equivalent requirements are

$$e_T + b + z\sqrt{Fv_*/M} \le \epsilon, \tag{34}$$

$$F_{\max}(b, r) = \frac{M(\epsilon - e_T - b)^2}{z^2 v_*}, \qquad b_{\max}(F, r) = \epsilon - e_T - z\sqrt{Fv_*/M}, \tag{35}$$

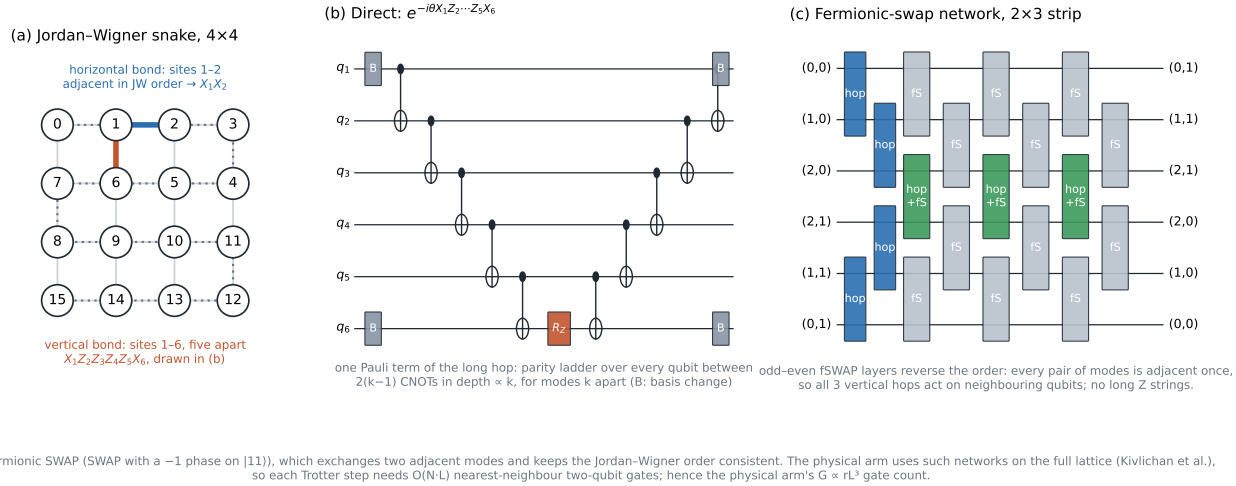


Figure 13: The physical circuit. (a) Jordan–Wigner snake on a 4×4 lattice: the horizontal bond 1–2 is adjacent in the ordering, while the vertical bond 1–6 carries a Pauli string across five positions. (b) Direct compilation of one Pauli term of that hop: a CNOT parity ladder across every qubit in between (B: basis change). (c) A fermionic-swap network on a two-row strip: alternating layers of fermionic swaps (fS) reverse the mode order, and each vertical hop is applied when its two modes meet, so all gates act on neighboring qubits.

where $z = 1.96$ and $e_T = C/r^2$. An exhausted allowance is infeasible. Figure 14 optimizes over a declared integer-step menu; the estimator must achieve its F and b at the selected r. We do not assume that its overhead is independent of refinement.

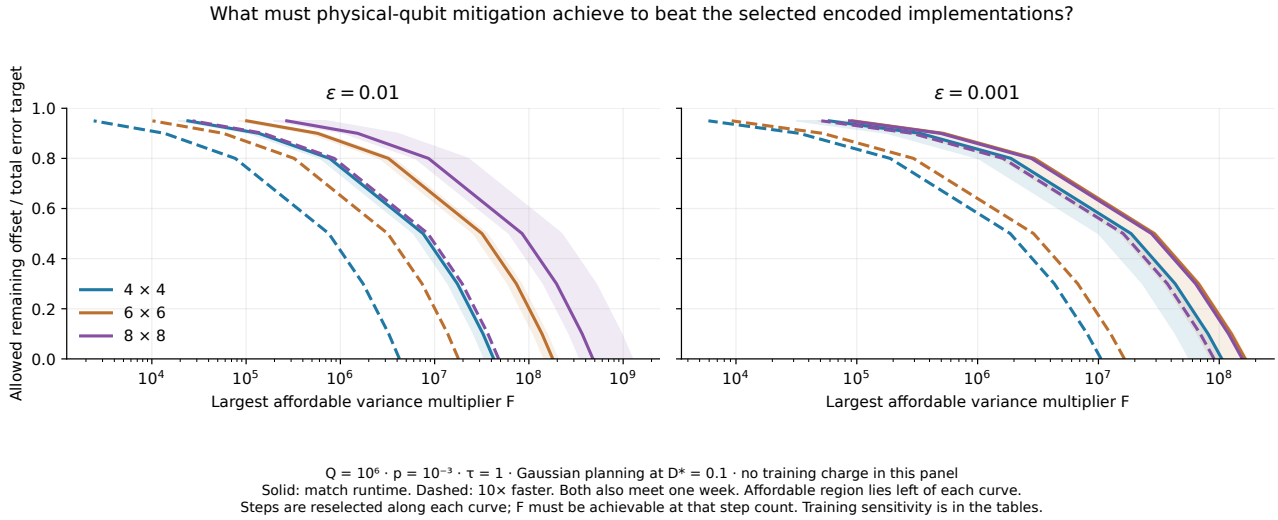


Figure 14: Requirements for physical-qubit computation at $Q = 10^6$, $p = 10^{-3}$, and $\tau = 1$. Each curve gives the largest affordable variance multiplier as a function of the allowed systematic offset. Solid curves match the fastest fitting member of the selected encoded implementations; dashed curves require a tenfold speedup. Both also meet the week cutoff. Bands span the encoded scenarios, with steps reselected along each curve. The panel assumes zero training charge; saved tables also charge six million training attempts. The affordable region lies to the left of a curve. These are Gaussian planning requirements on an unknown estimator, not predictions of an experimental pipeline's accuracy.

For a requested speedup s, we set $T = \min(T_{\text{week}}, T_{\text{encoded}}/s)$, with $s = 1$ or 10. The encoded comparator is the fastest fitting member of the three selected implementations within that scenario. Those implementations were optimized for one-week area, not reoptimized over all allocations for runtime at this particular Q. If no encoded copy fits, only the one-week physical requirement is reported. Training uses zero or six million additional equal-duration attempts as explicit sensitivities. A real pipeline must also account for uncertainty in its trained correction and any shared time-series information.

For example, at 4×4, $Q = 10^6$, and $p = 10^{-3}$, allowing a remaining offset $b = 0.001$ at $\epsilon = 0.01$ gives $F_{\max} = 3.3 \times 10^7$ at 13 steps, with a 67-second encoded comparator. At $\epsilon = 0.001$, allowing $b = 10^{-4}$ gives $F_{\max} = 8.1 \times 10^7$ at 40 steps, with a 14-hour comparator. These central-scenario examples charge no training. The larger affordable multiplier in the strict case follows from the longer encoded runtime; it does not make physical accuracy easier. An actual estimator must attain both the stated offset and variance at the corresponding circuit depth.

The scientific decision is whether plausible accuracy and sampling costs fall on one side of these boundaries. A detailed mitigation-transfer calculation is most useful where resolving that question could change the ranking. We apply the same distinction between measured components and assumed composition to all three architectures. A capacity leader is called robust only when its worst capacity across this scenario menu exceeds the best capacity of both other encoded arms. Overlaps remain unresolved, and this criterion establishes no unconditional physical-qubit size boundary.

F QUOPS model details

Specification. The benchmark definition follows the paper's Supplementary Information: CNOT layers first, partial final layers, the embedded polarization $\gamma = (4^w F_e - 1)/(4^w - 1)$ excluding terminal state preparation and measurement, the one-sided 95% pass rule, and the score and rate formulas. Our model does not include the confidence margin or the SPAM reference circuits; with 10^4 executions the statistical margin is small. The complete extraction, including every measured row and 27 comparisons with another workshop implementation, is in `docs/QUOPS_SPEC.md`.

Polarization. Exact process polarizations of random circuits with $w = 2$–8 and $p = 10^{-3}$–10^{-2}, computed by Pauli-transfer and independent Choi calculations, agree with Eq. (1) to 0.215%. The product of local gate polarizations underestimates them by factors 1.07–1.24 at mean polarization near 0.2, because a local polarization does not account for the register dimension. Idle time lowers the polarization by 4–11% at odd widths 3–7.

Routing. We routed 900 random circuits of widths 4–144 on the smallest near-square grid with Cirq's RouteCQC and with a constructive network of pair assignment, edge colouring, and row–column–row odd–even sorting. Per benchmark layer for $16 \le w \le 144$, RouteCQC needs $0.071w^{1.68}$ added SWAPs and $0.35w^{1.59}$ fused two-qubit gates, with two-qubit depth growing as $w^{1.04}$; the constructive network needs about three times more gates but its depth grows only as $w^{0.62}$. Doubling the available area changed no cost under the tested placement. The model uses the measured means through $w = 144$, the fits beyond, and the router with smaller polarization loss at each width.

Encoded arms. Each layer of long-range CNOTs is assumed to take two code clocks by lattice surgery in the fast block, without congestion. The isolated-gadget calibration of Appendix B supports the clock count and the memory law for adjacent CNOTs; congestion at large widths remains the least tested assumption. STAR averages its injection instrument over the wrapped angles with sixteen

samples. FT splits the loss budget between synthesis, magic states, and logical faults on a declared grid and optimizes ports, factories up to 4096, cleanup, and distance.

G Fixed-time benchmark

Figure 15 repeats the capacity comparison at fixed $\tau = 1$. Beyond $L \approx 8$ it compares the cost of evolving full lattices rather than the accuracy of a local answer, which is why the main text leads with $\tau = L/4$. The ordering is the same, with larger analog gains: at $p = 10^{-3}$, $\epsilon = 0.01$, and 10^8 qubits, STAR reaches $L = 12$ and the hybrid $L = 20$, compared with 10 and 16 under the reference menu.

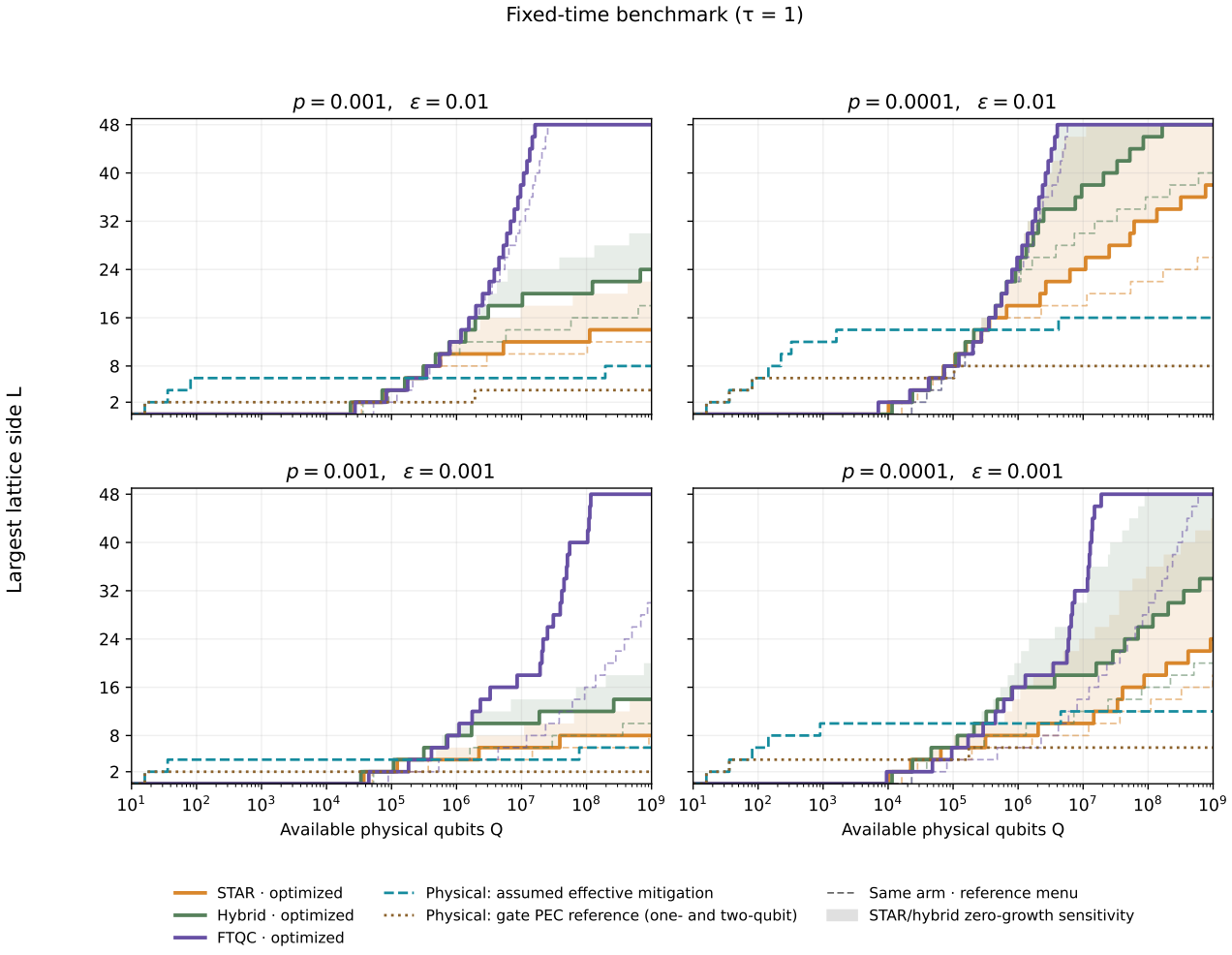


Figure 15: Fixed-time benchmark at $\tau = 1$ under uniform noise with cultivated T states. Solid curves use every optimizer option and thin dashed curves the reference menu; bands give the STAR and hybrid zero-growth sensitivity. Beyond $L \approx 8$ these curves compare the cost of evolving full lattices, not the accuracy of a local answer.

H The design search

Every capacity in this paper is the answer to one optimization problem. For a workload instance (a lattice L with $\tau = L/4$, or a random-circuit shape (w, s)), a physical error rate p, a precision ϵ, and an architecture, we find the fewest physical qubits $Q_{\min}$ that complete the task within the week while meeting the accuracy contract of Eq. (3). The capacity at a budget Q is then the largest instance with $Q_{\min} \le Q$. This appendix describes what the search ranges over, why each option is in the menu, which inputs are declared rather than chosen, and where the calibrated constants come from.

Objective and constraints. A candidate design fixes a sampler, one complete copy of the computation with area q_a and attempt time t_a, and an allocation of the error budget. It is admissible when every bias fits its share of ϵ, when one attempt fits in the week, and when the statistical share leaves a finite shot count $N_0 e^{\chi}$. Its cost is $Q = q_a \lceil N_0 e^{\chi} t_a / B \rceil$ [Eq. (4)], or the Richardson-weighted form of Eq. (11). Every architecture pays this same accounting, so the search compares designs on one currency: physical qubits for one week.

Decision variables. Table 7 lists the menu. Each entry is a choice a quantum architect would make, and each trades area against time or error. The menus start at the reference design of our earlier integrated model and extend well past it in the direction that the trade favors, so that the reference design is always one point in the search.

Declared inputs. Some inputs are hardware or engineering assumptions rather than choices, and the search holds them fixed: the 240 ns syndrome cycle, 10 ns gates, 100 ns measurement and reset, and 200 ns feedforward; the tile charges per lane (6) and per extra port (2); two code clocks per lattice-surgery CNOT layer; the transfer of a cultivated state into the data distance in $2\max(d, 15)$ cycles; the sampling convention $D_* = 0.1$ and $z = 1.96$; and the effective physical mitigation overhead κ. Each declared input that could move a conclusion has its own sensitivity run (Sec. 5.7). On all-to-all hardware the delivery charges change as described in the hardware sections, and nothing else.

Calibrated constants. The formulas that cost a design contain constants measured in separate calculations, never fitted to the capacities:

- the surface-code law A, Λ from 9.6×10^7 Stim memory shots per noise model (Appendix B);
- the Trotter coefficients $C(\tau)$ and K_4 from exact dynamics on patches up to 4×4 and strips up to 6×3 (Appendix A);
- STAR's accepted channel at every workload angle from exact enumeration of generator faults, and its growth error from 2.4×10^7 decoded growth shots (Appendix C);
- synthesis T counts from the gridsynth law $3.02 \log_2(1/\zeta) + 0.77$, checked against compiled CUDA-Q words (Sec. 5.4);
- cultivated-state error and service time from Ref. [6], and factory designs re-priced with our surface-code law (Appendix B);
- Pinnacle's codes, logical-error law, and magic engines from Ref. [22];
- for random circuits, routing laws measured on 900 routed circuits and the embedded polarization product validated against exact simulation (Appendix F).

Variable	Menu	Arms	What it trades
Trotter steps r	$\lceil\sqrt{C/(f\epsilon)}\rceil$ for 21 Trotter shares $f = 0.03$–0.97, at least 8τ; then ± 2 around the best	all	Trotter error against exposure (analog, physical) or T count (FT)
Extrapolation	single branch, or Richardson with $(r, 2r)$ and weights $(-\frac{1}{3}, \frac{4}{3})$	all	a K_4/r^4 residual against ≈ 4.9 times the sampling time
Error split	synthesis 0.01–$0.15\,\epsilon$; logical 0.005–$0.2\,\epsilon$; magic by source; residual bias 0.1ϵ (analog, physical) or 0 (FT); statistics the rest	encoded	where the error budget is best spent
Code distance d	every odd d from 3, smallest admissible	encoded	area against logical-fault bias
Data layout	compact (one port, 9 clocks per T) or fast (1 clock per T)	STAR, hybrid, FT	area against T-consumption speed
T-consumption ports	1–128 (fast layout)	FT, hybrid	parallel T consumption against 2 tiles per extra port
Factories	1–1024	FT, hybrid	supply rate against source area
T source	cultivation, with or without a 15-to-1 cleanup; distillation factories (24–34 designs per p)	FT, hybrid	output error against area and service time
Equal-angle batching	batches of 1–1024 by Hamming-weight phasing	FT, Pinnacle	fewer synthesized rotations against Toffolis and workspace
Analog lanes	1–128, at most L (termwise) or N (plaquette)	STAR, hybrid	parallel injection against 6 tiles per lane
Injection seed / receiver	$d_i \in \{5, 9, 25\}$, receiver 13 or 25	STAR, hybrid	preparation acceptance and error against growth error
Retry scheme	raw repeat-until-success, or noisy PCEC	STAR, hybrid	half the dephasing against a cancelled coherent term
Pinnacle code / engine	five GB codes ($d = 4$–24); two magic engines per p	Pinnacle	block area and cycle length against logical error; engine area against output error

Table 7: The menu of the design search for the Hubbard workload. The random-circuit search uses the same variables with lanes, ports, and factories up to 1024–4096, four synthesis shares, and, in the mitigated mode, a menu of allowed losses.

How the search runs. The search is exhaustive over the product of the menus. For each combination of discrete choices, the code distance is scanned upward and the first distance that meets the logical budget is taken, since the area only grows beyond it. The remaining choices are evaluated together as arrays, so that a single full-fault-tolerance instance at $L = 16$ evaluates about 10^7 candidate designs. Two rules prune without changing the answer: analog designs with exposure $\chi > 120$ are dropped, because they would need more than 10^{24} qubits under any attempt time; and noisy PCEC is evaluated only where raw repeat-until-success stays below that cutoff, since PCEC never has less dephasing. Among admissible designs the search keeps the one with the fewest qubits, breaking ties by the shorter attempt. The search is deterministic and reproducible from the saved inputs.

What the answer means. The result is the optimum over the menu, not over every conceivable design. Three checks bound the difference. The reference menu alone reproduces our earlier integrated model in 335 of 336 replayed cases. Each option added one at a time changes the leader in only a few of 64 cells (Table 3), and together in 15. The sensitivity rounds move each declared charge by factors of two to four and change few crossovers. None of this replaces a compiled schedule: the costs are those of the declared delivery models evaluated at their best settings.

I Reproducibility

Every number in the main text is regenerated from saved results by the repository's `Makefile`. The uniform-noise results are rounds v4 and v5 of `python/optimization_round.py` (`pbs/optimization_round.pbs` with `ROUND=v4` or `v5`), the QUOPS study run with `QUOPS_CONFIG=configs/quops_sd6.json`, the surface-code and growth calibrations of `python/surface_law_sd6.py` and `python/growth_sd6.py`, and the factory menu of `python/factory_calibration.py` with the uniform-noise fit. `make audit` replays every saved calculation against its declared inputs. The earlier noise model's results (rounds v1–v3, `results/quops_v1`, and `results/surface_law_v1`) are retained. The Supplementary Material preserves the development history, earlier reference models, and compilation checks.

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